Géométrie hyperbolique
ckage luahyperbolic

cité de Lorraine, IECL)

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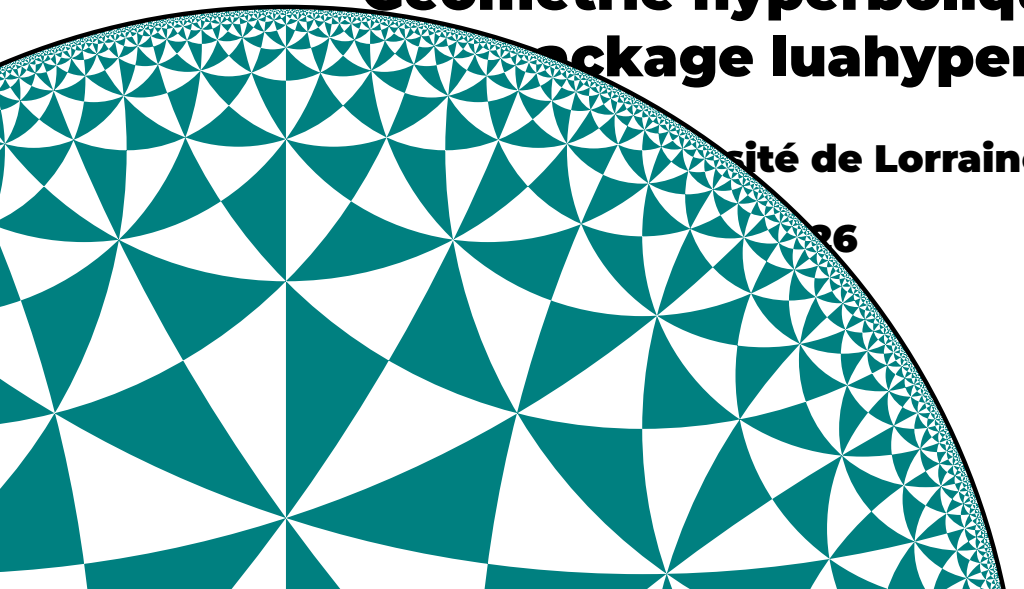

Plan :

1. Géométrie hyperbolique
2. Lua et LuaLaTeX
3. Le package luahyperbolic

Partie I
Introduction à la géométrie hyperbolique

Geometries of Surfaces
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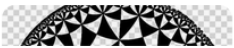

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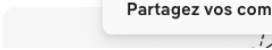

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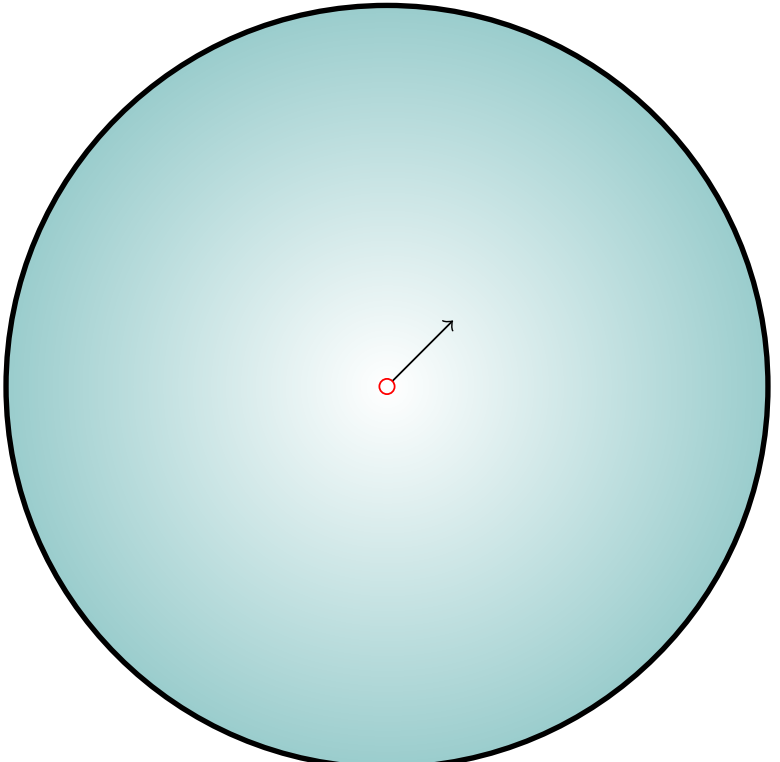
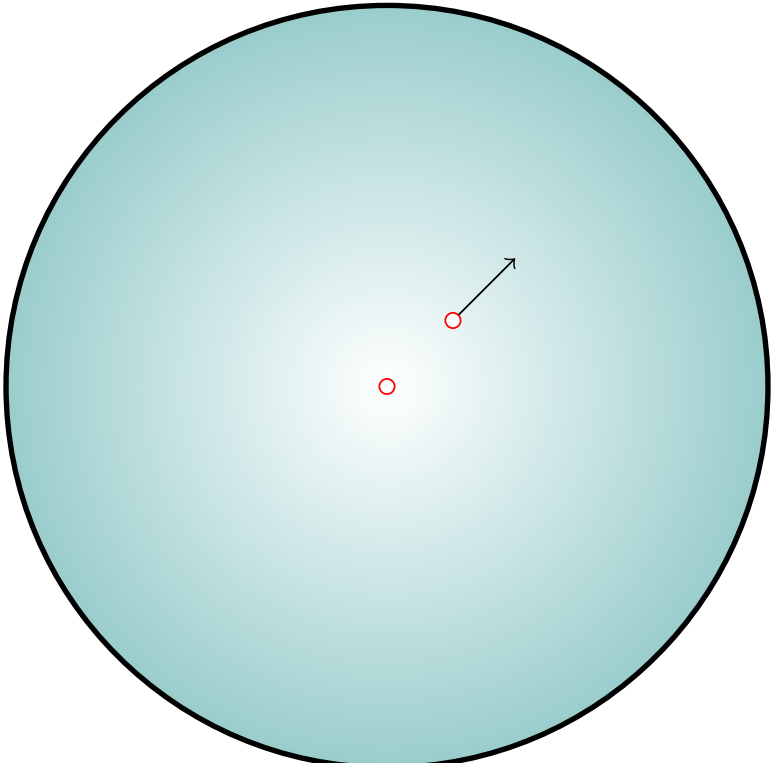
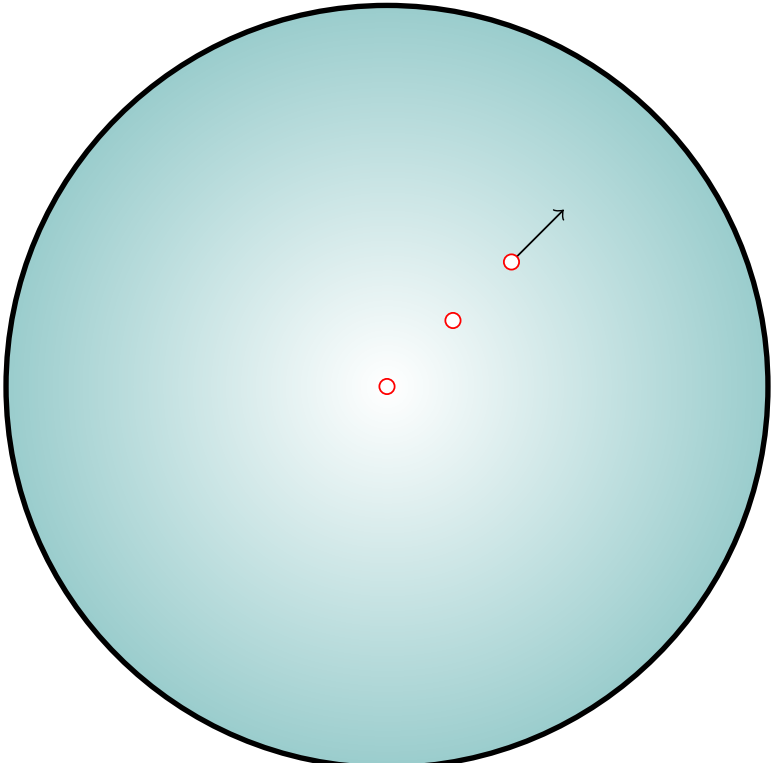
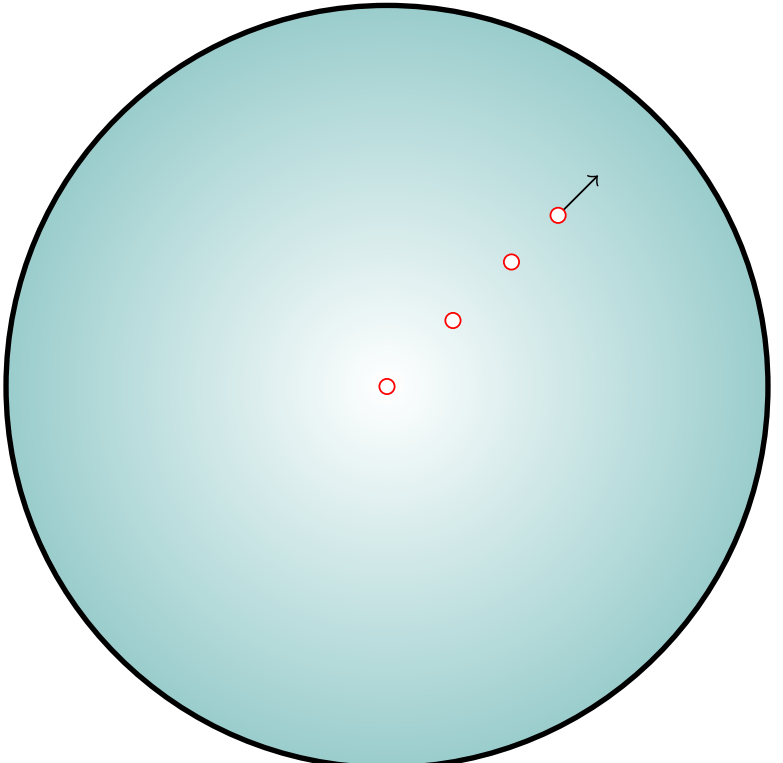
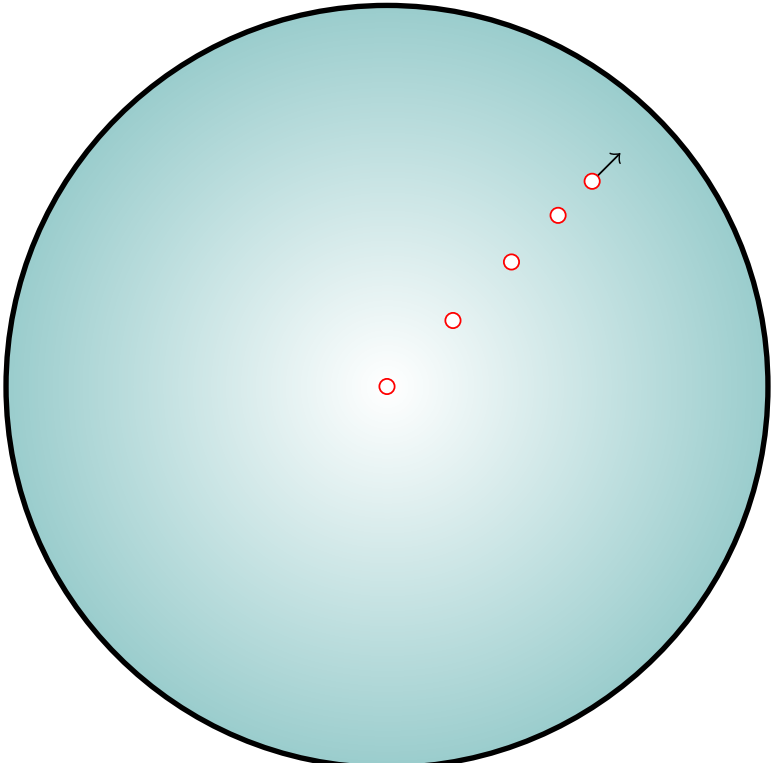
Le modèle du disque de Poincaré

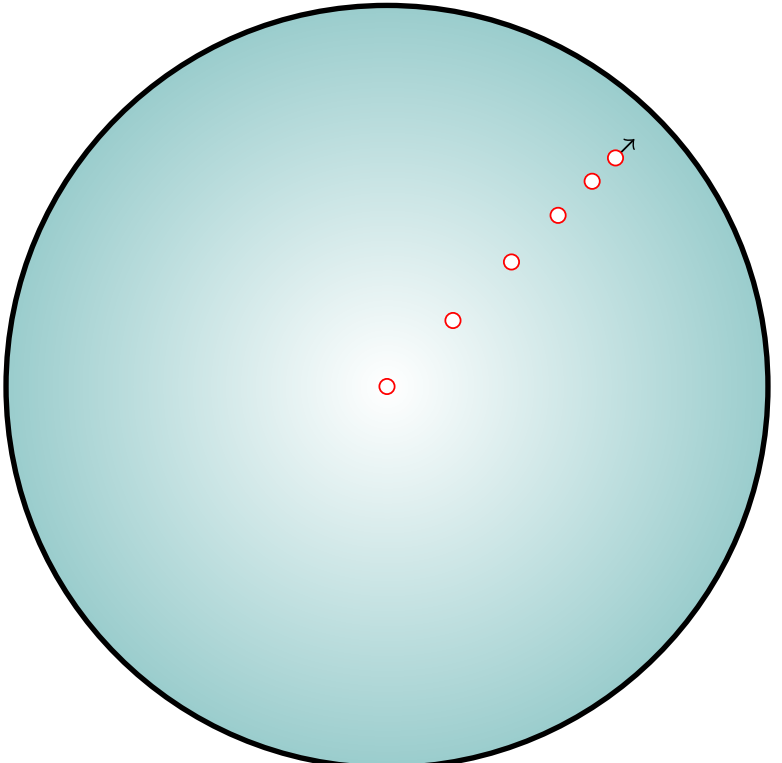


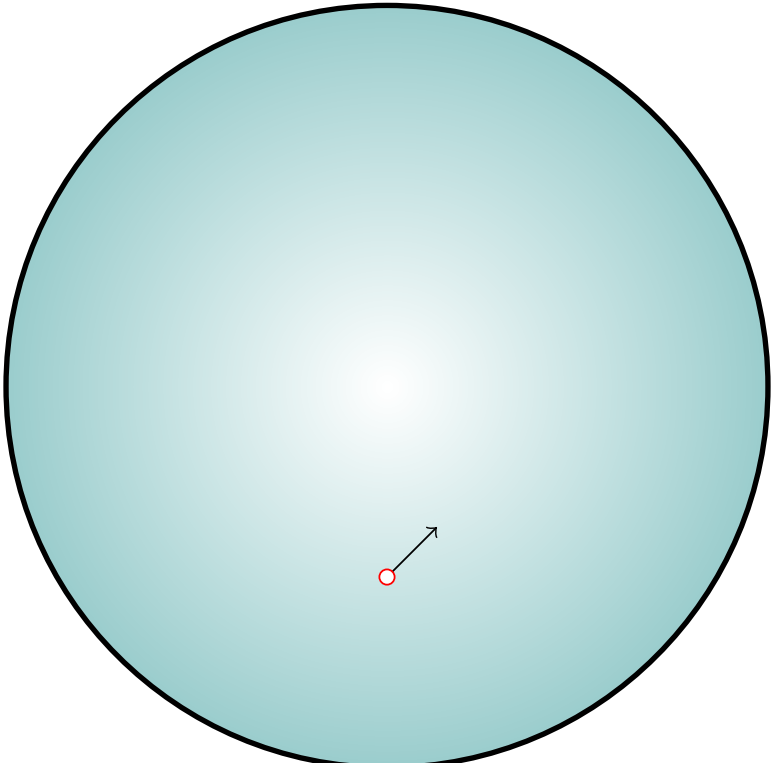


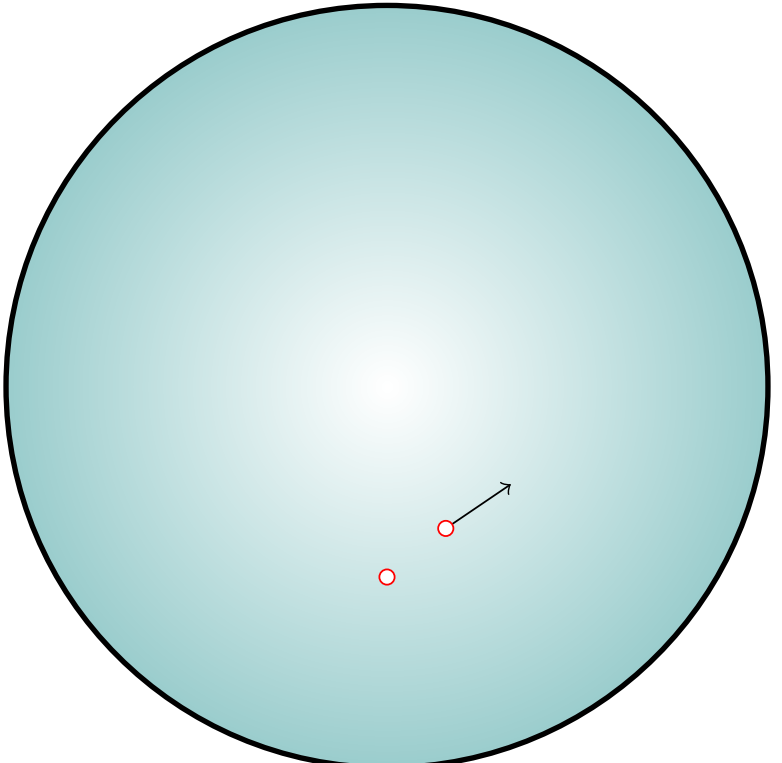


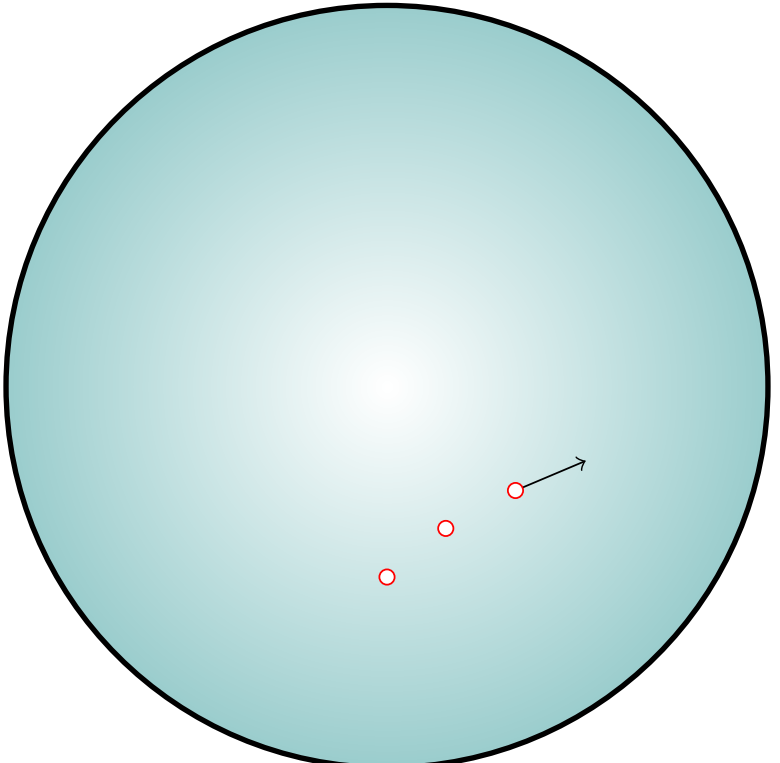


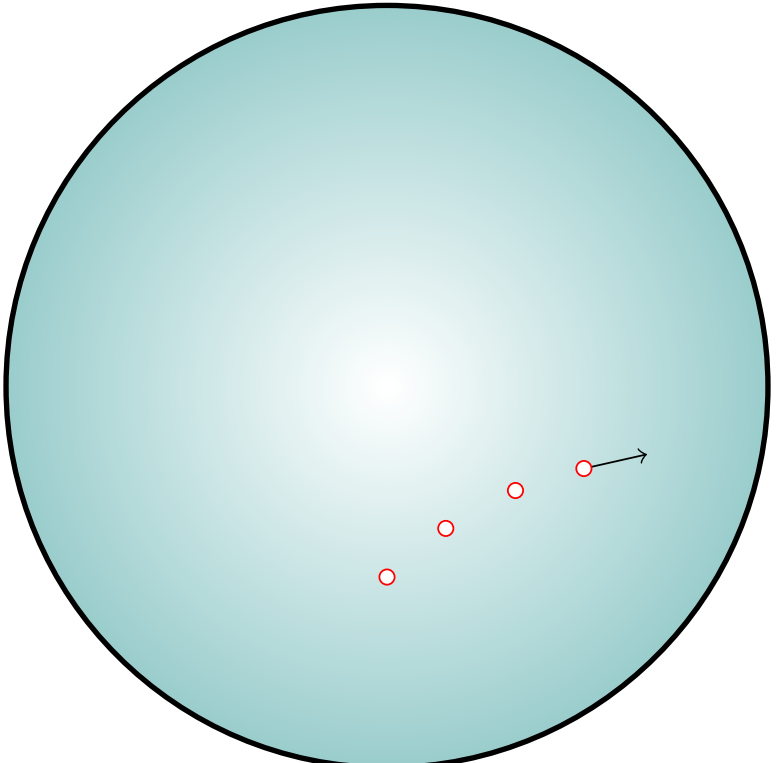


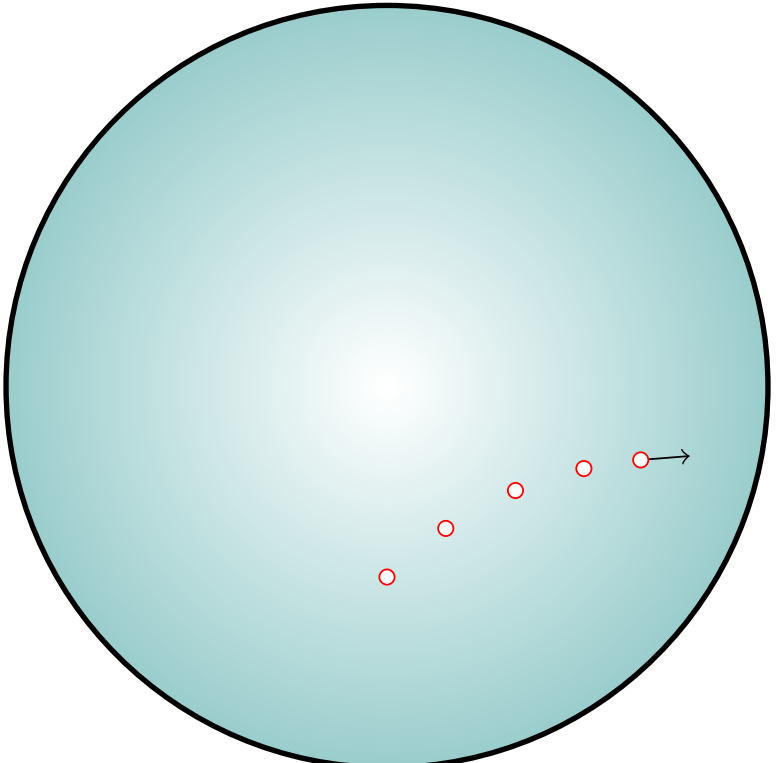


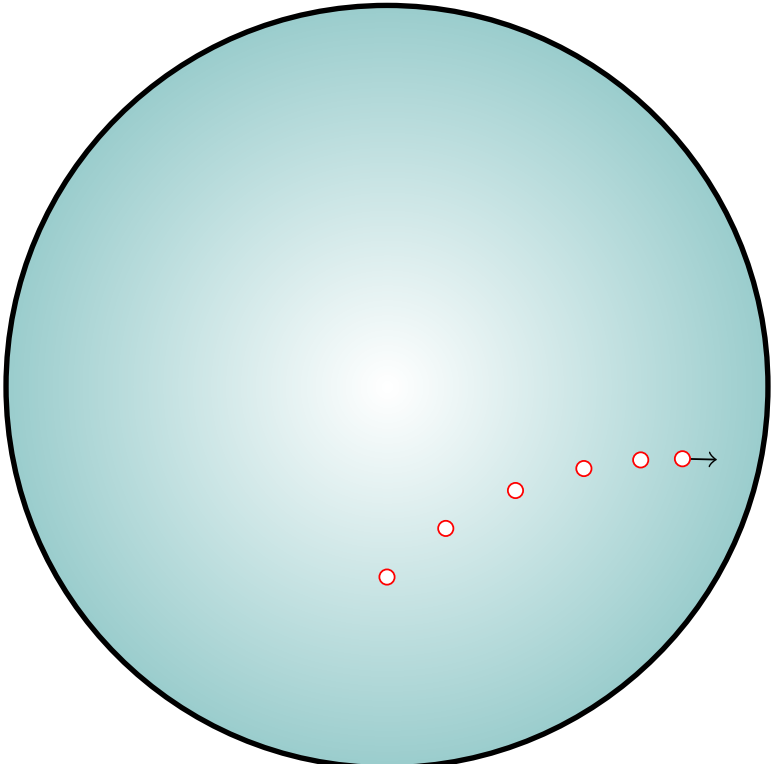




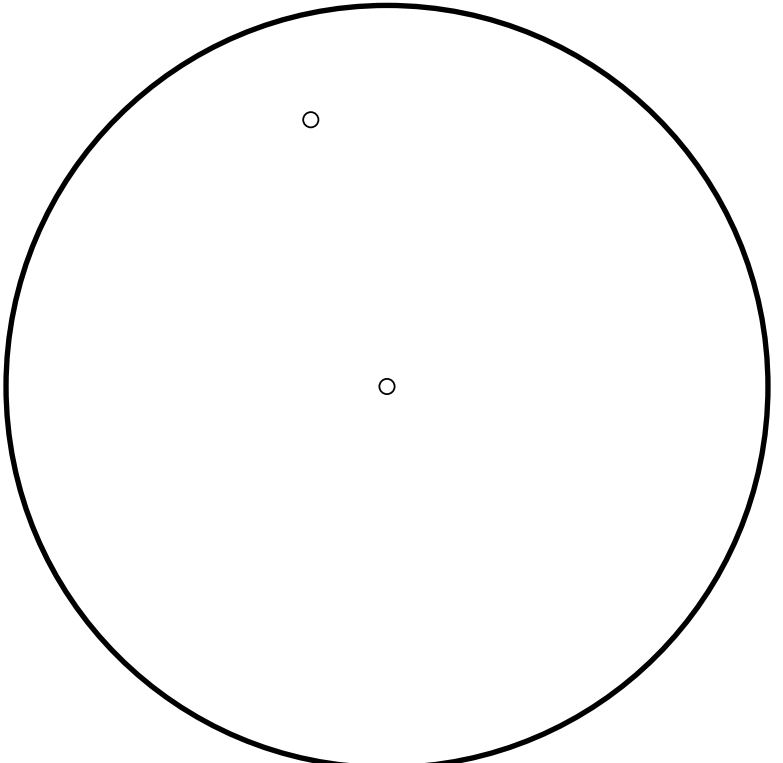


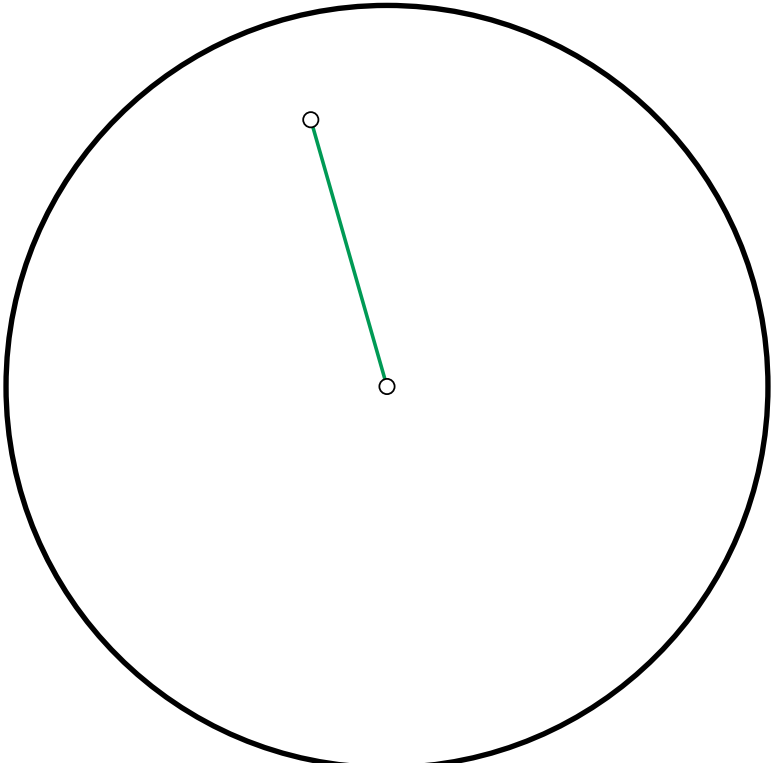


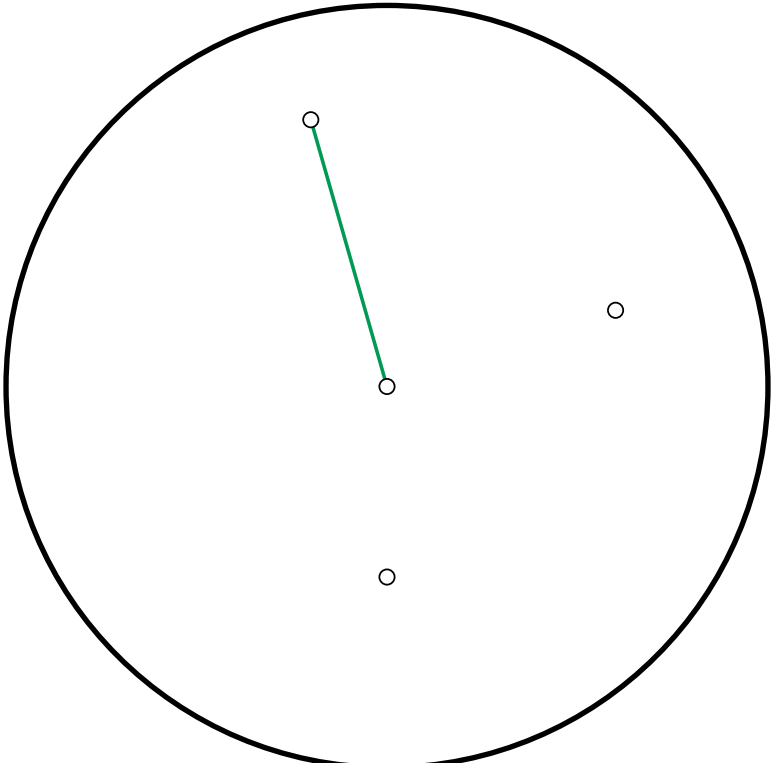


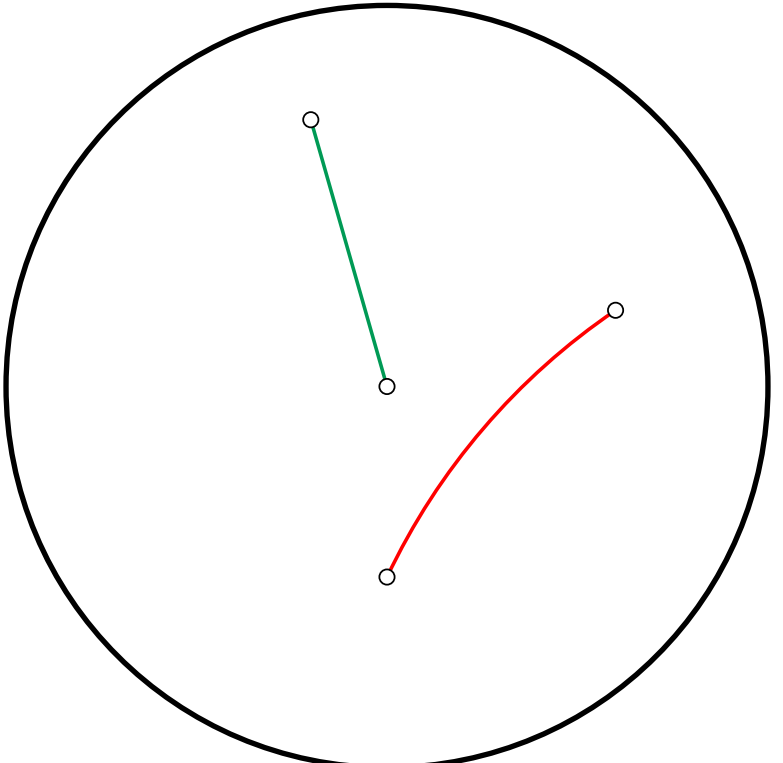


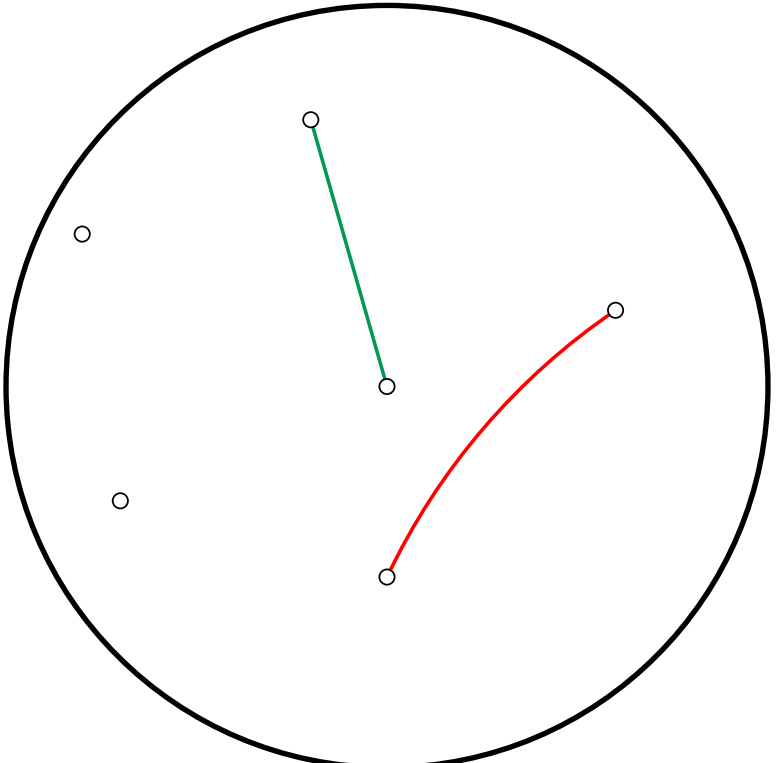
Géodésiques, segments et droites

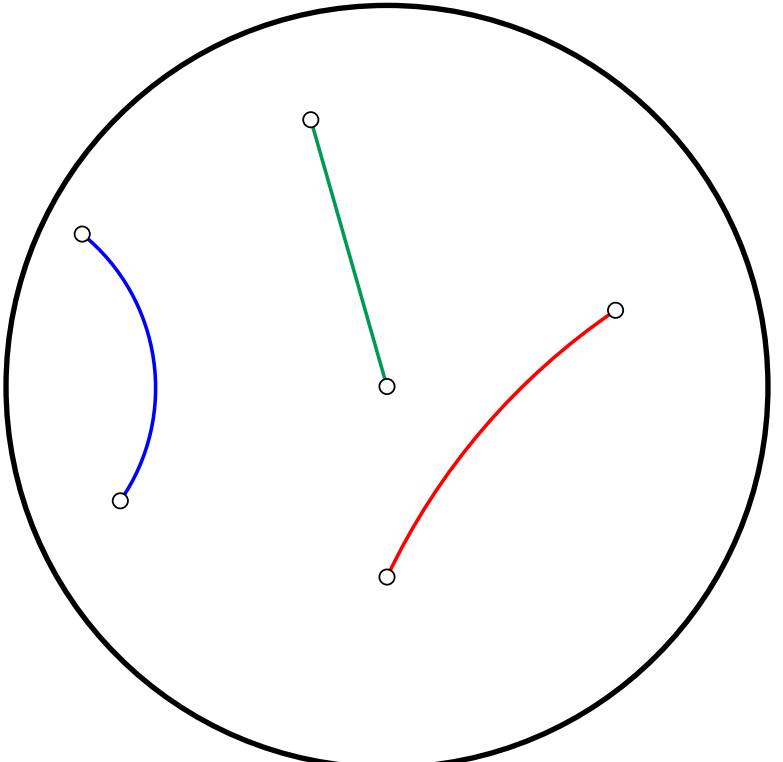


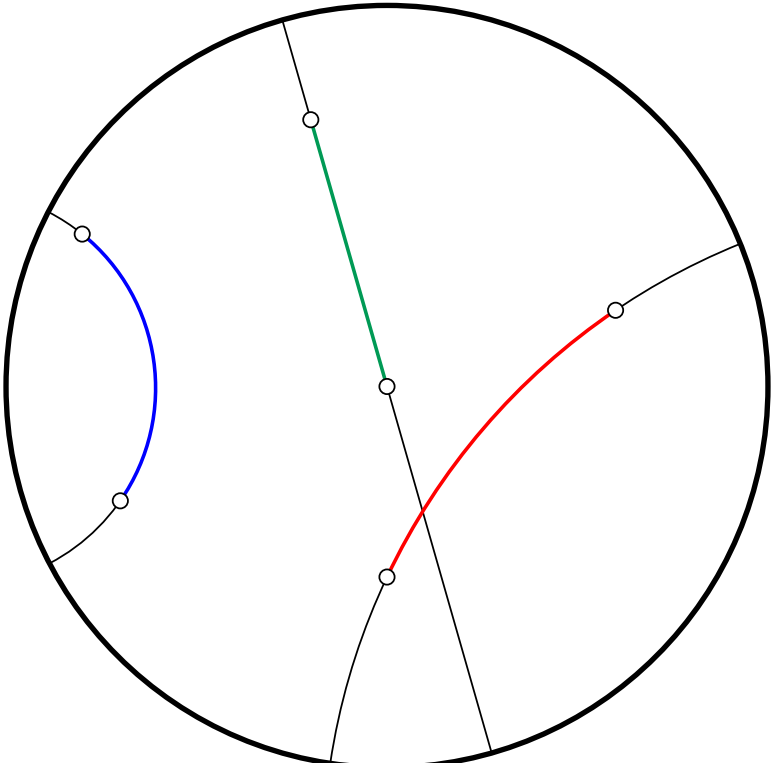




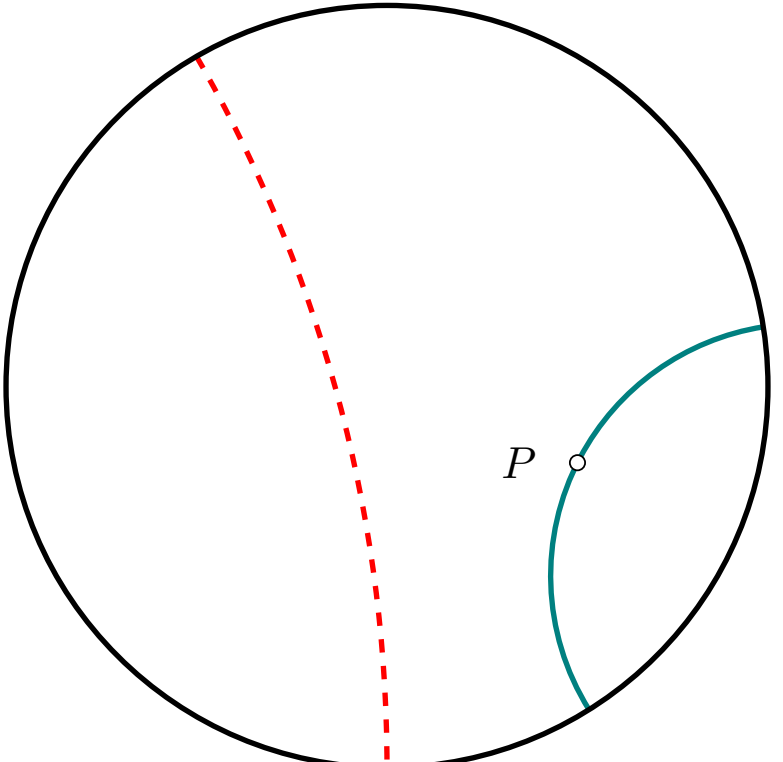








Attention !!
Le 5ème postulat d'Euclide tombe en défaut !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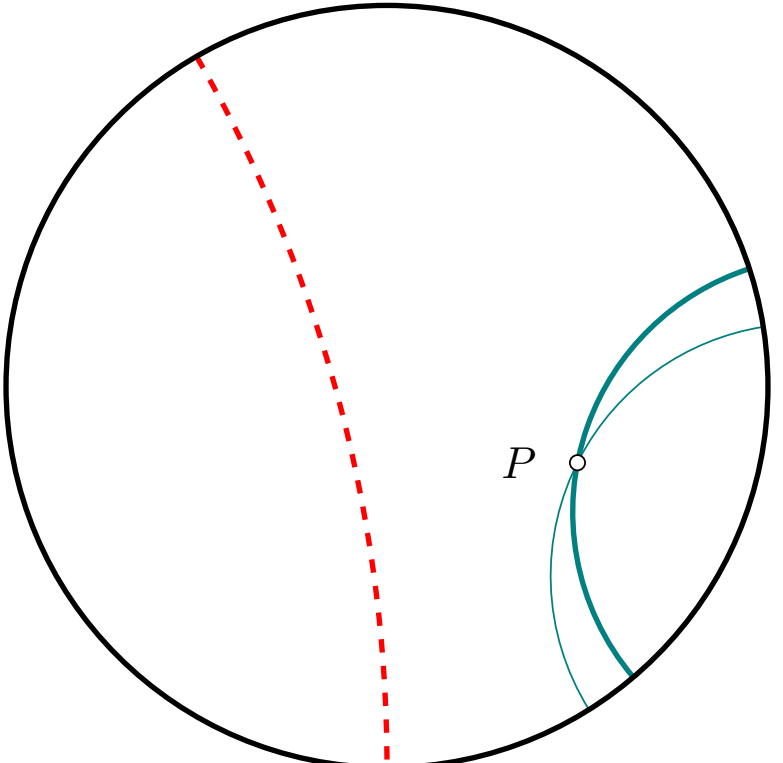
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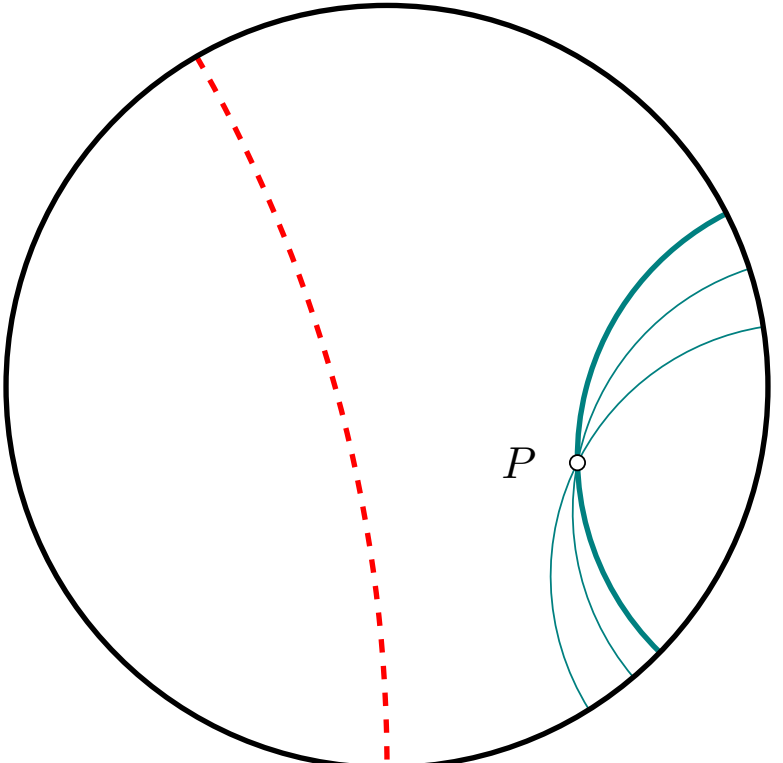
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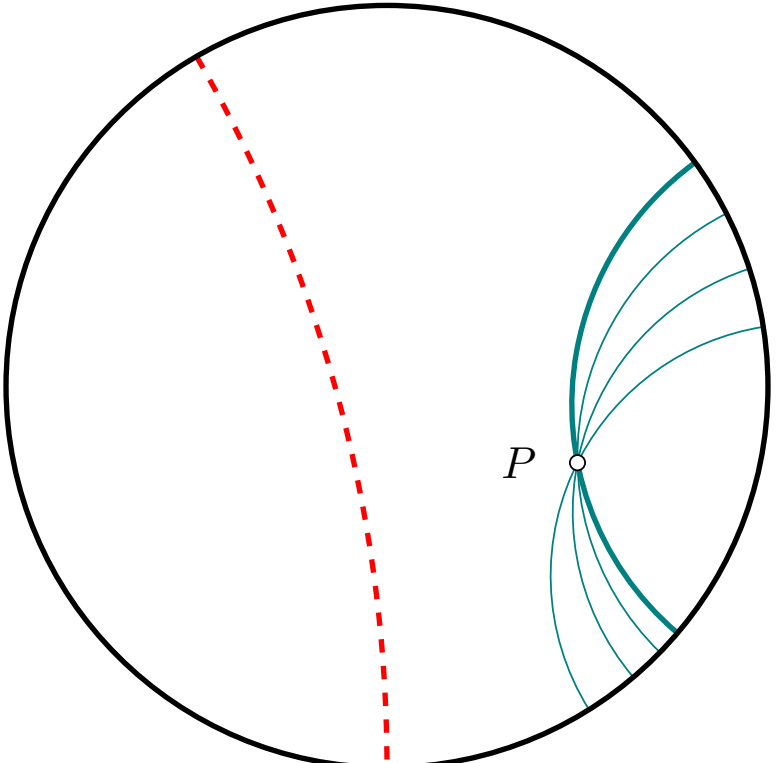
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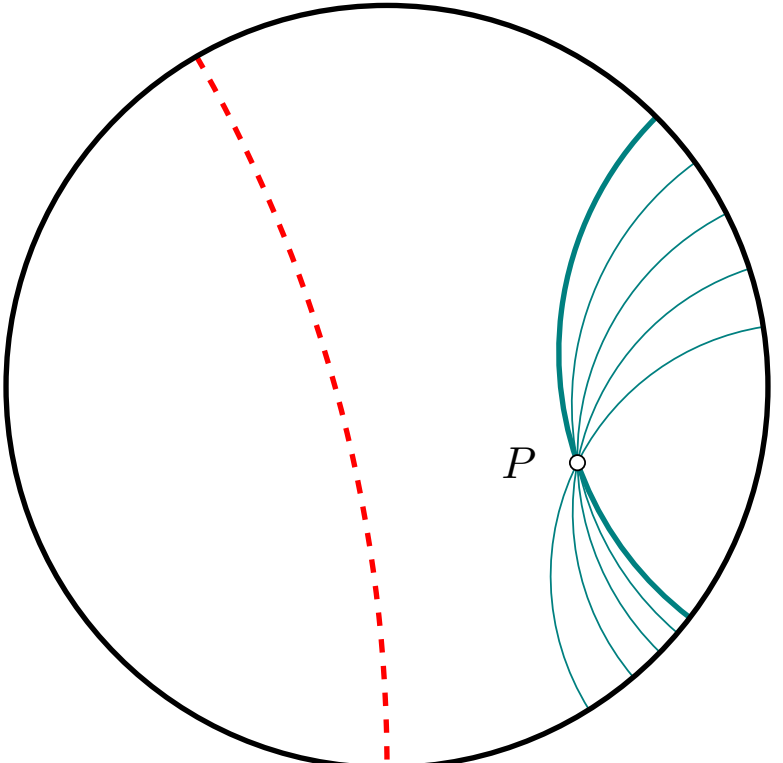
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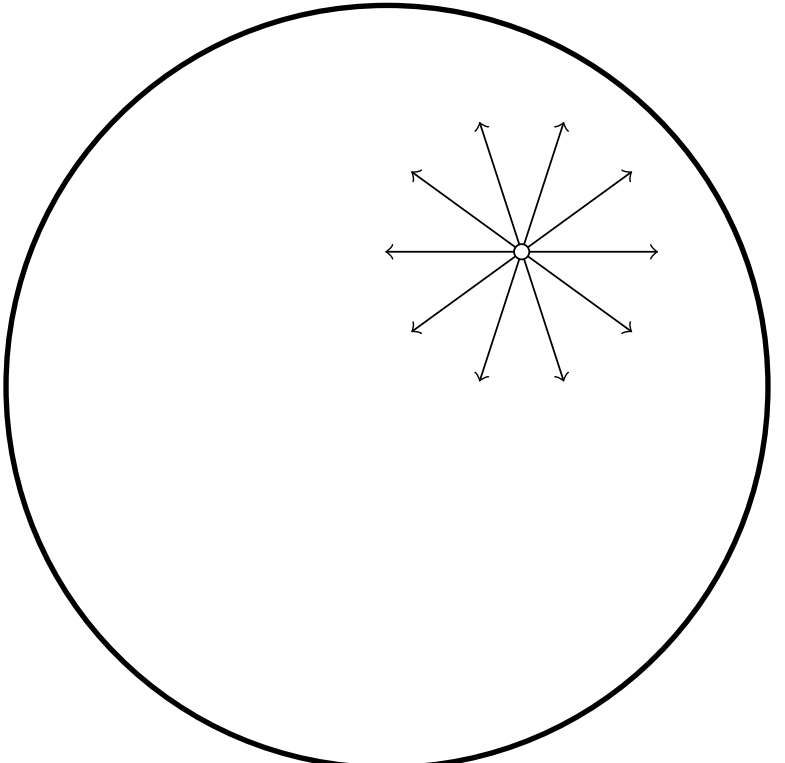
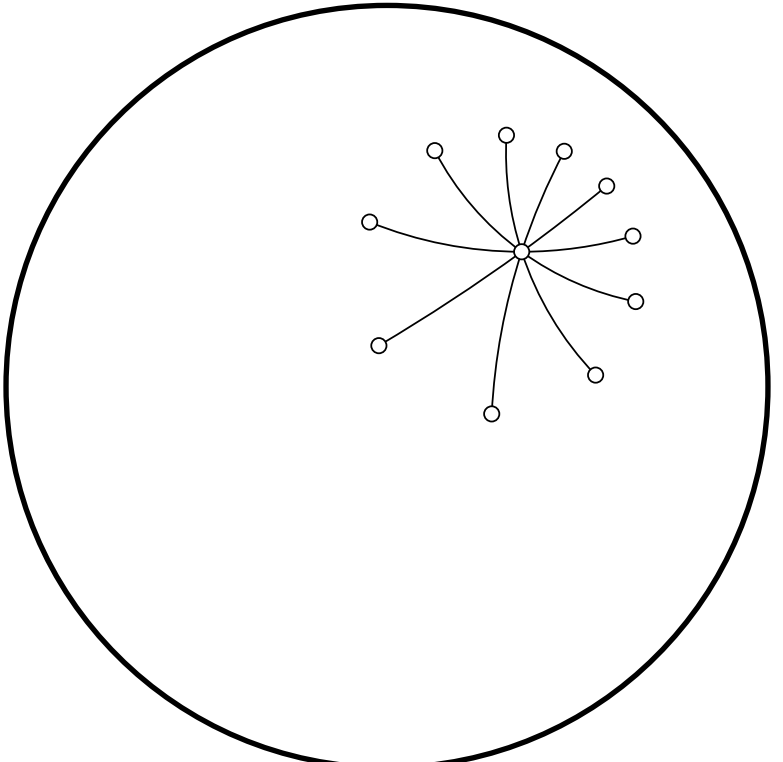
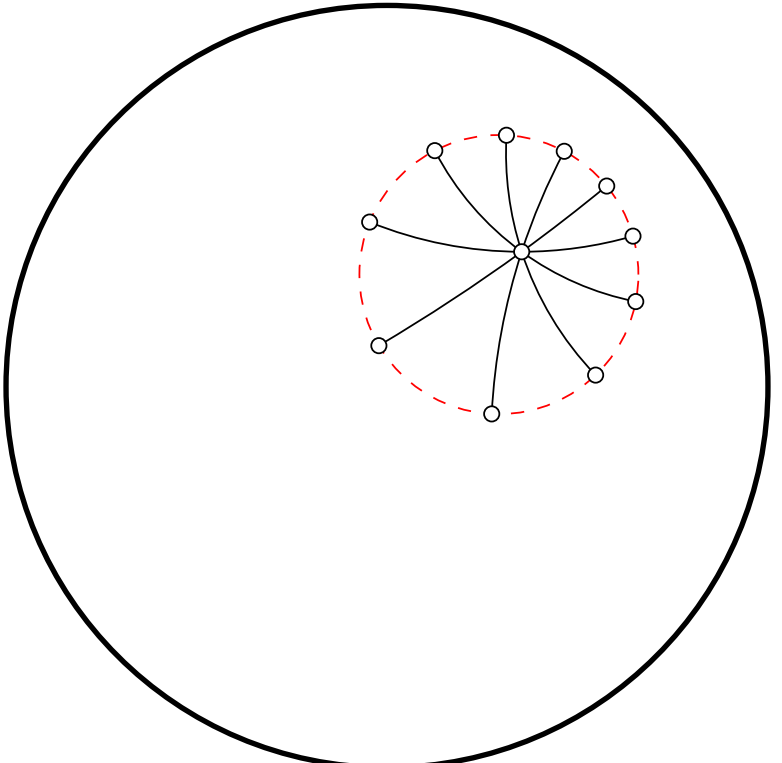
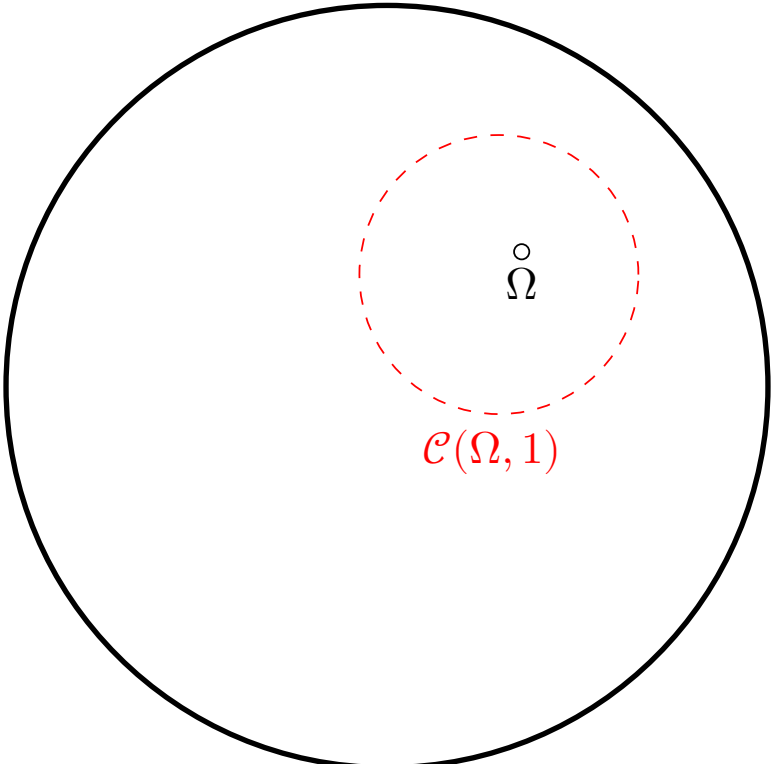
Cercles





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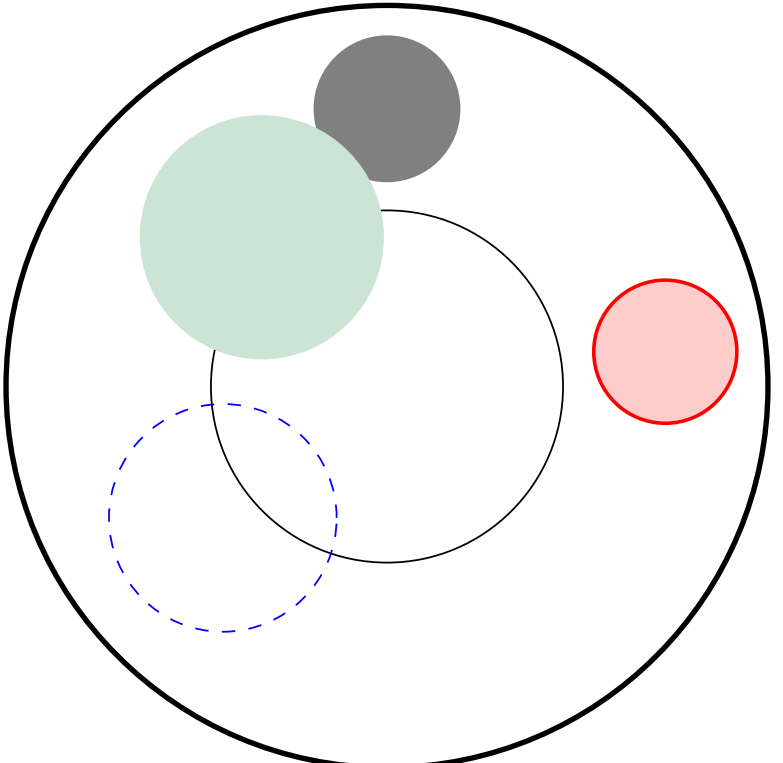

Triangles :
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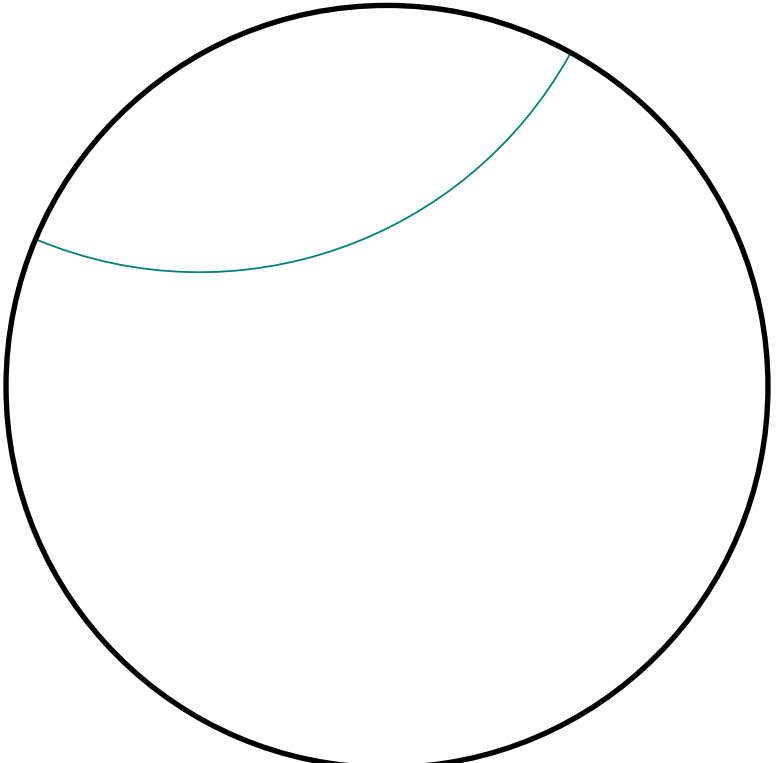

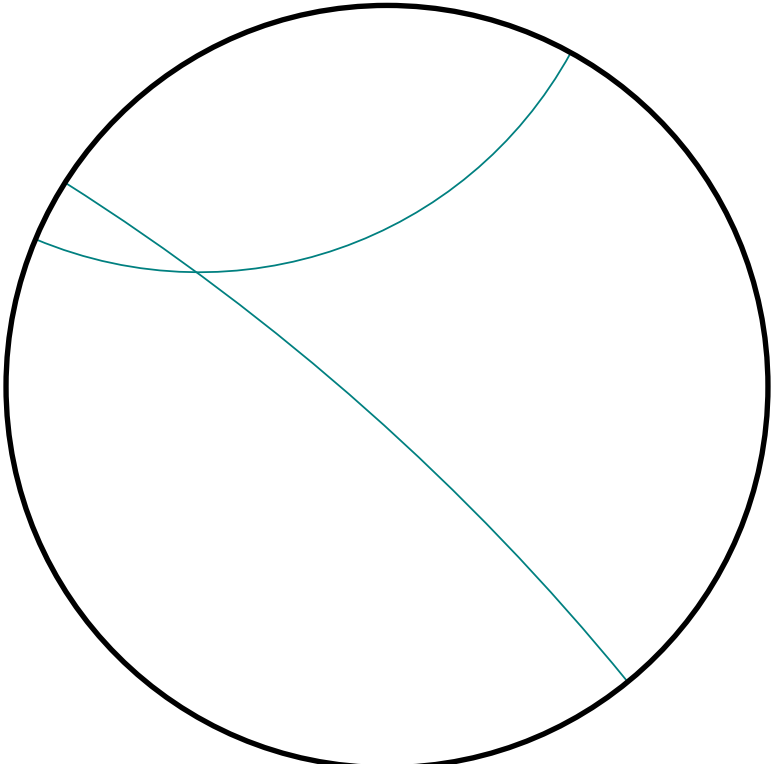

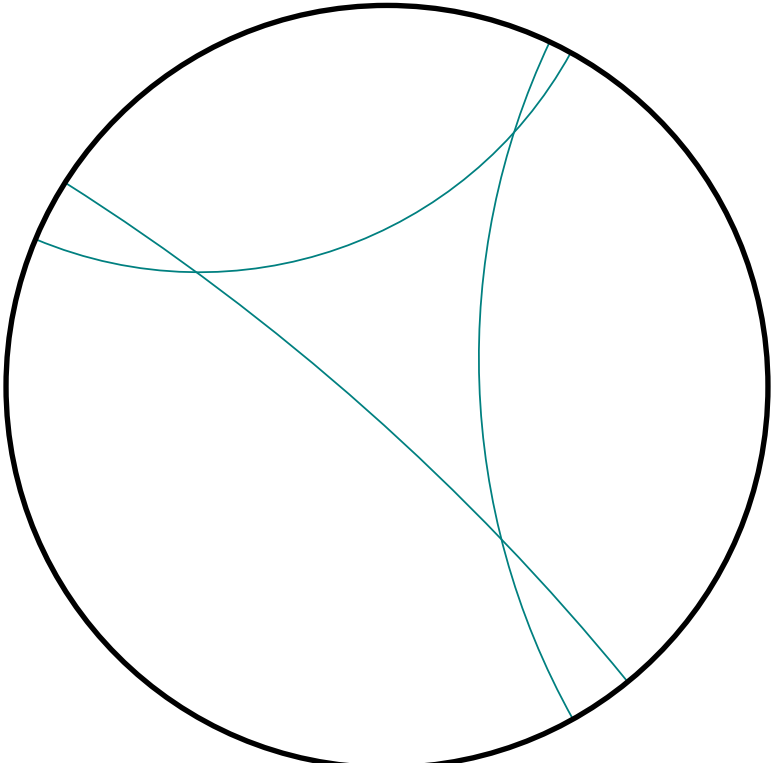

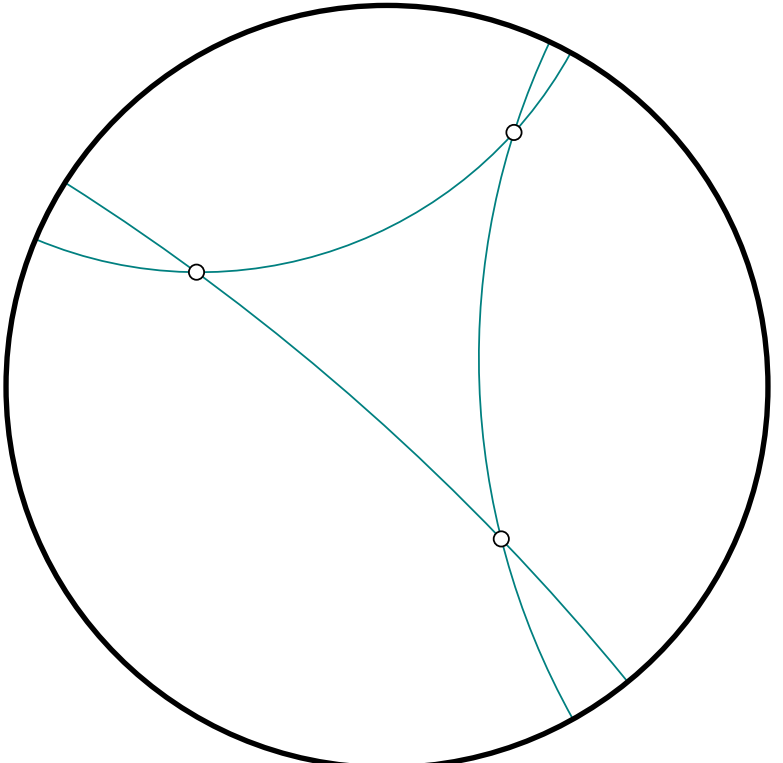

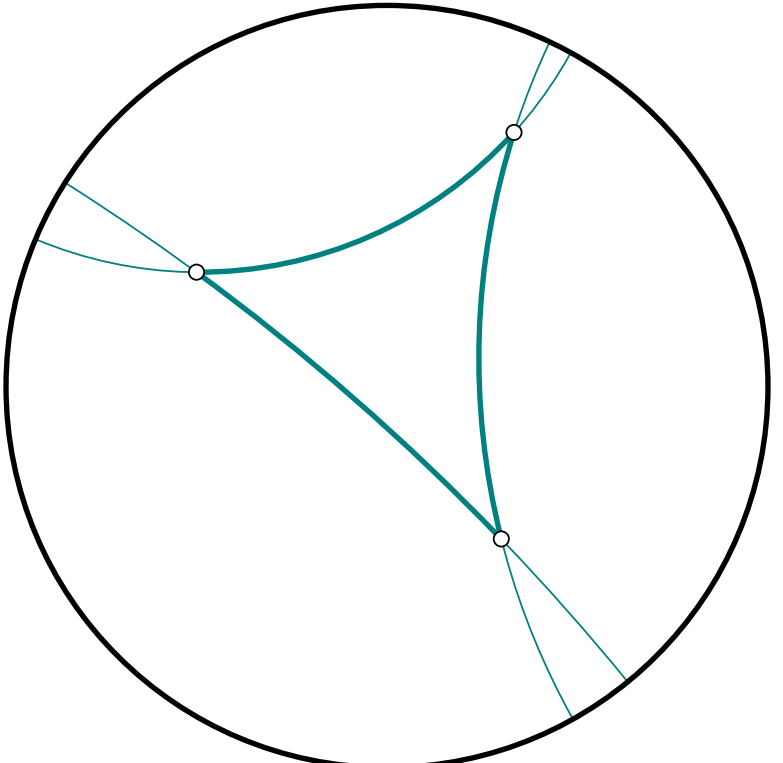

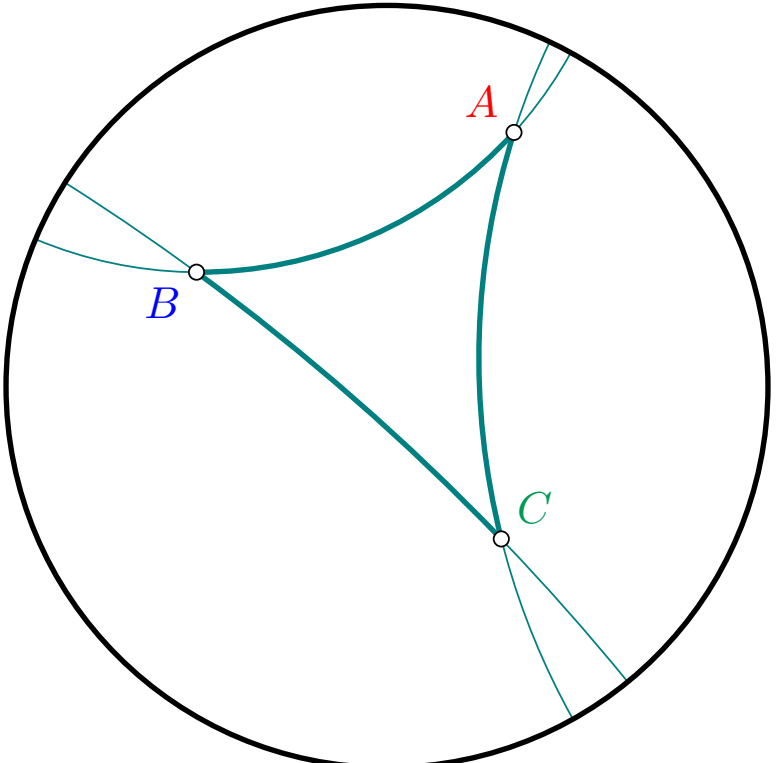
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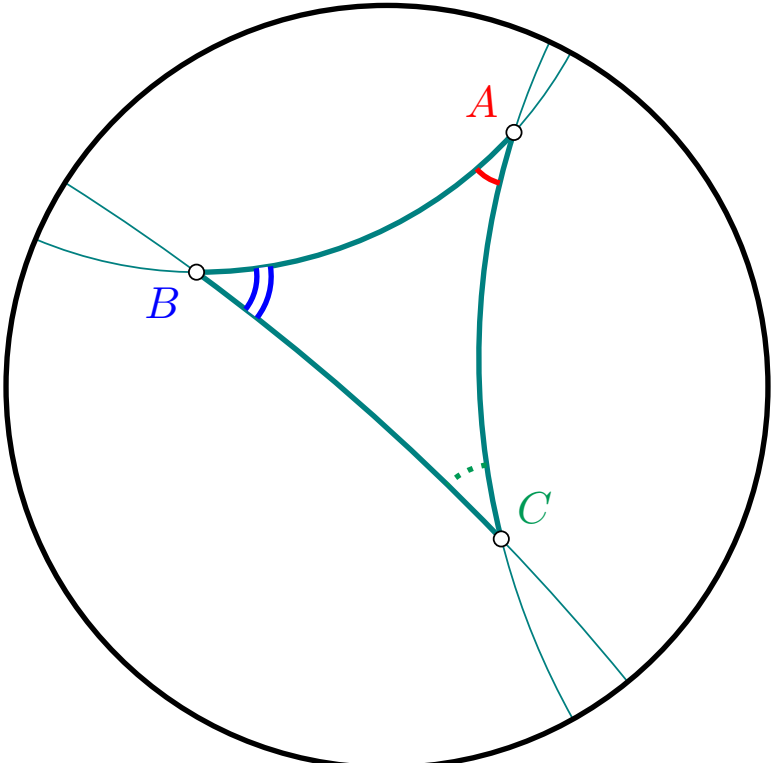
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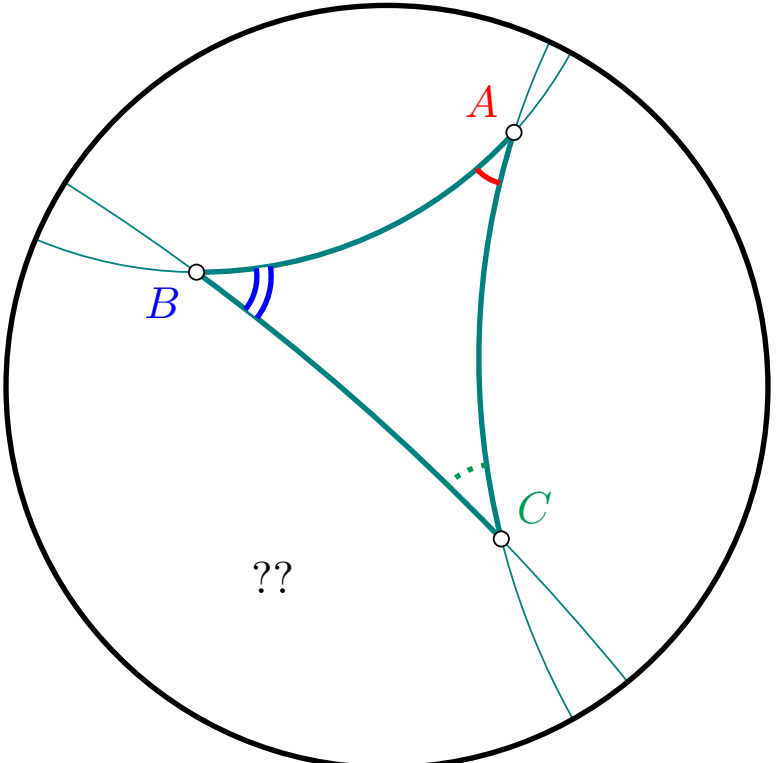
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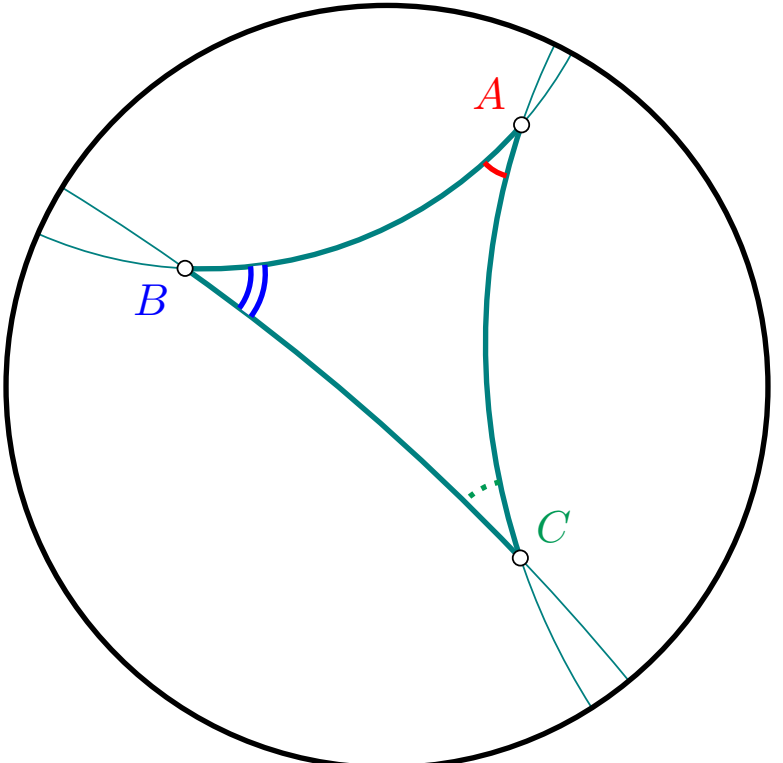
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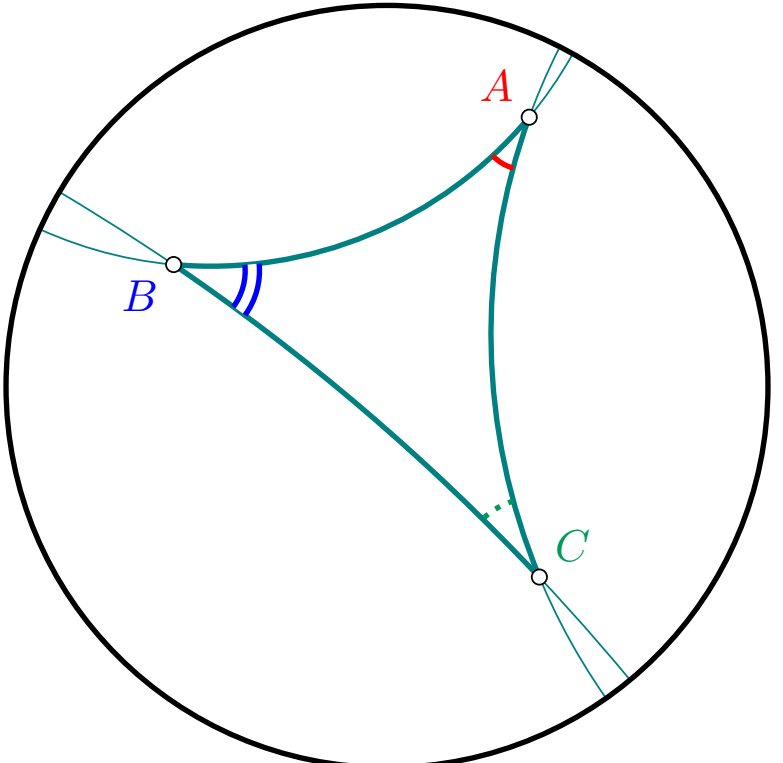
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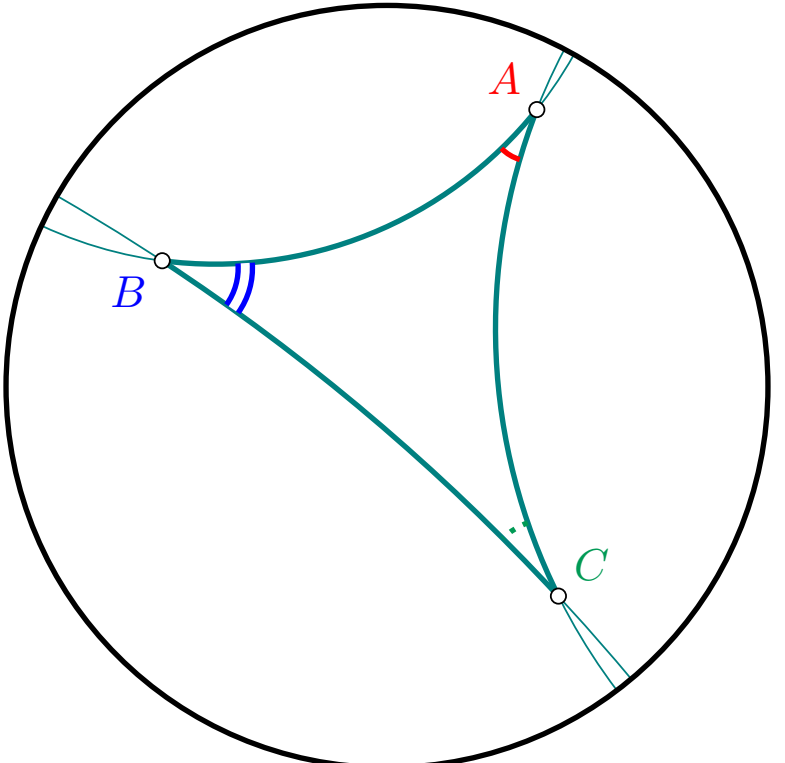
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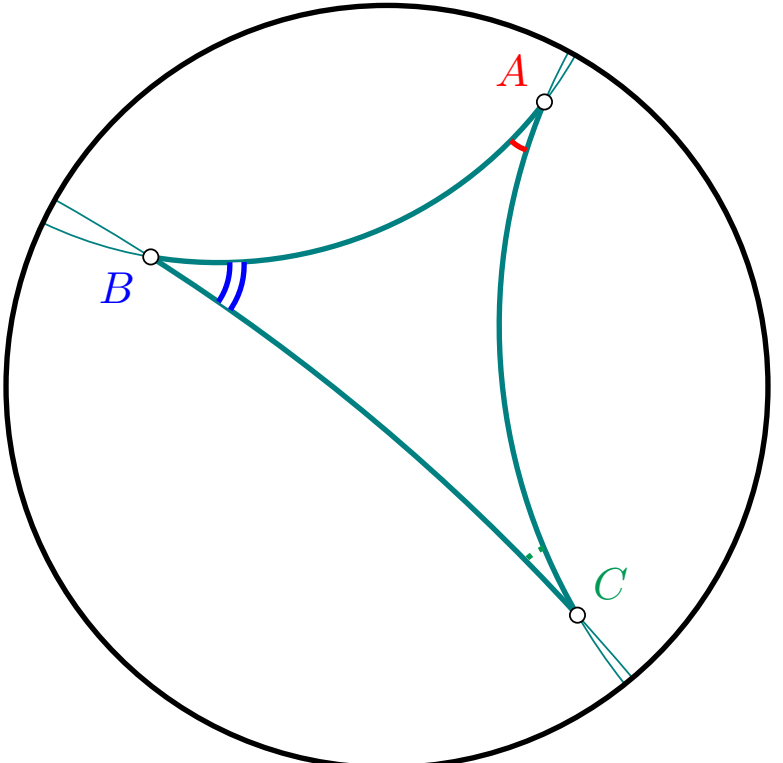
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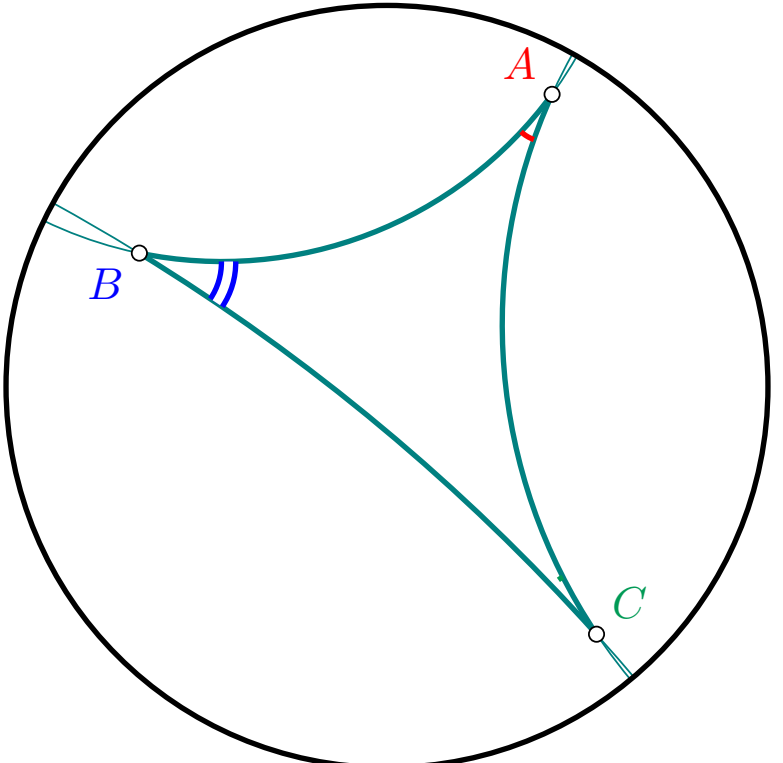
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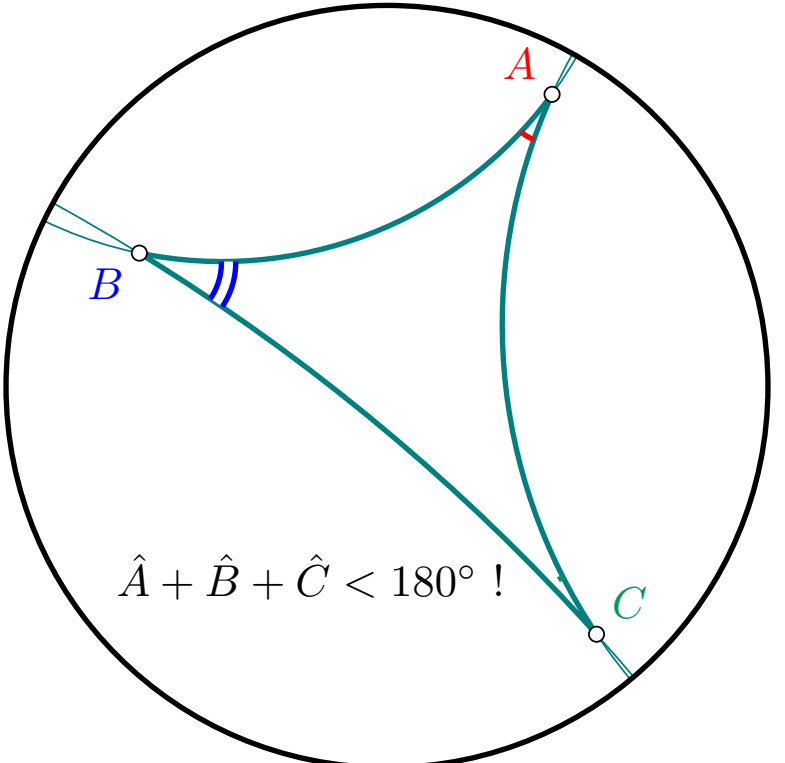
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Triangles :
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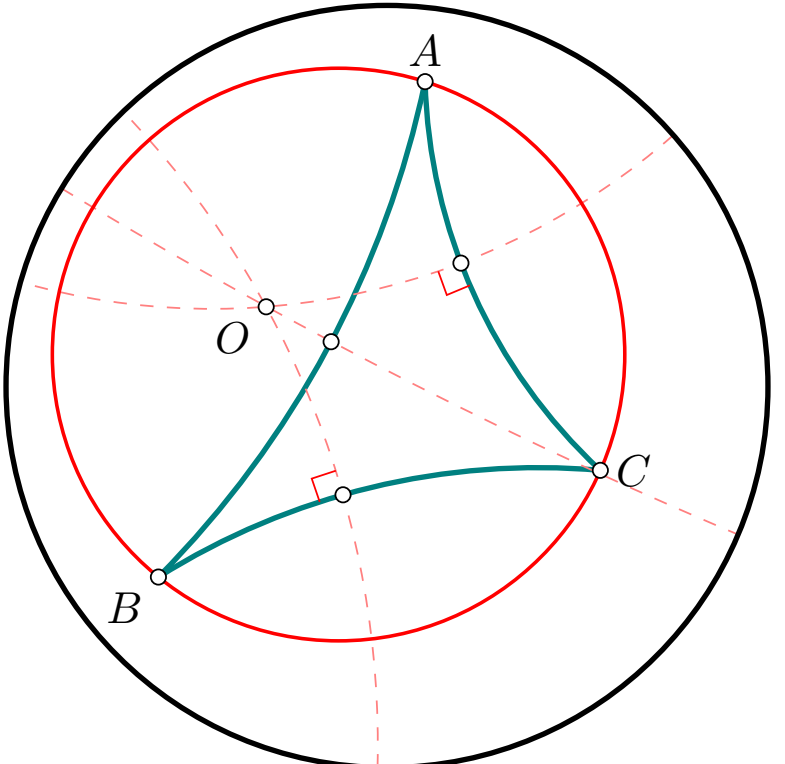
(Cercles circonscrits)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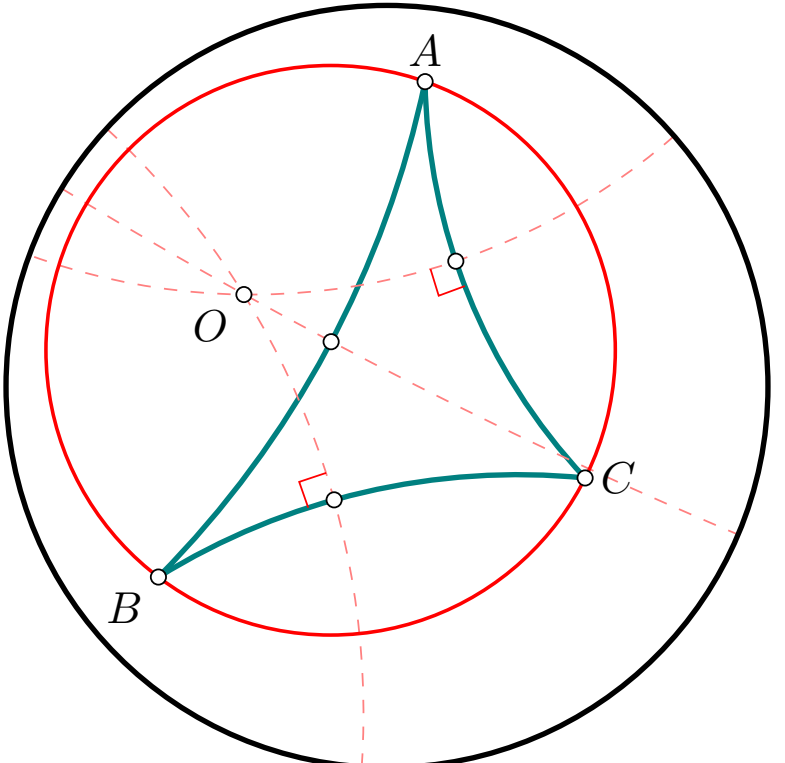
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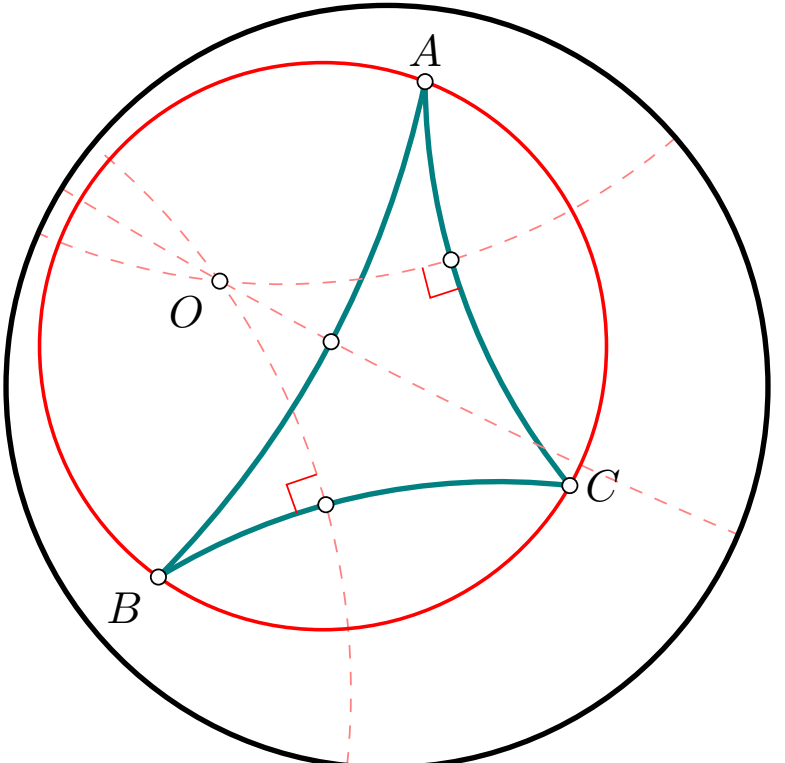
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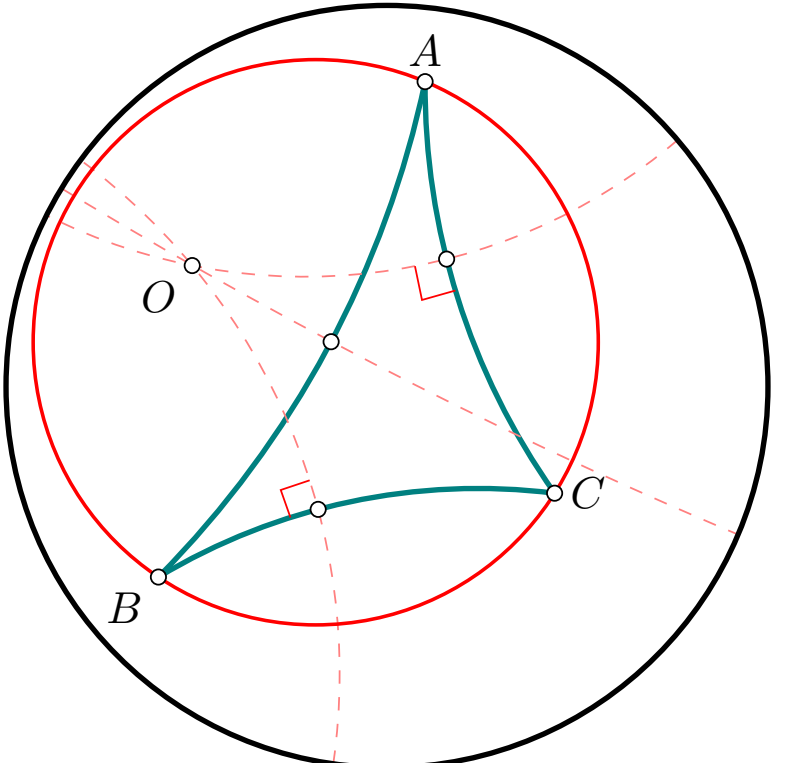
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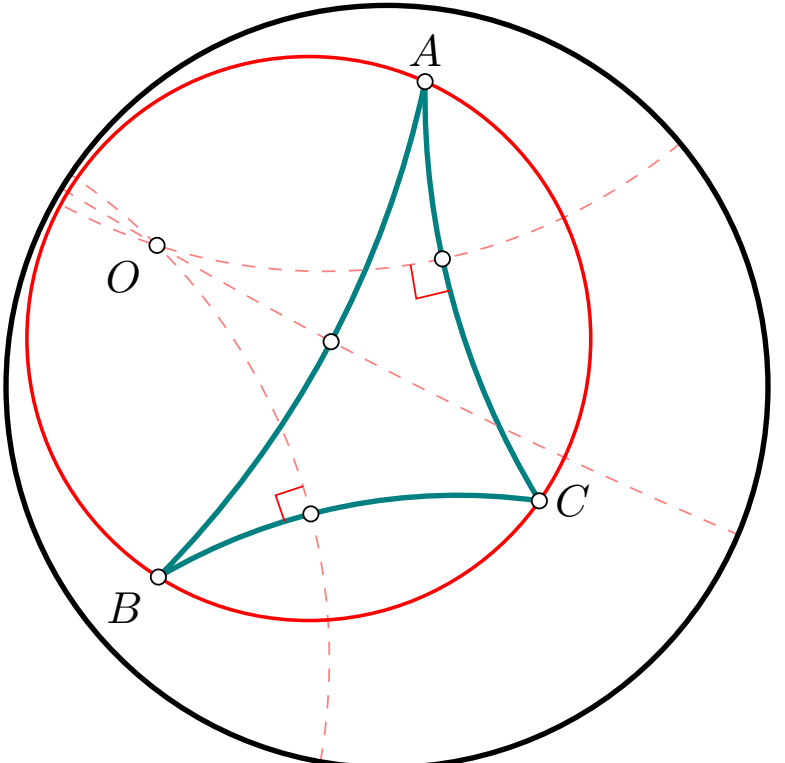
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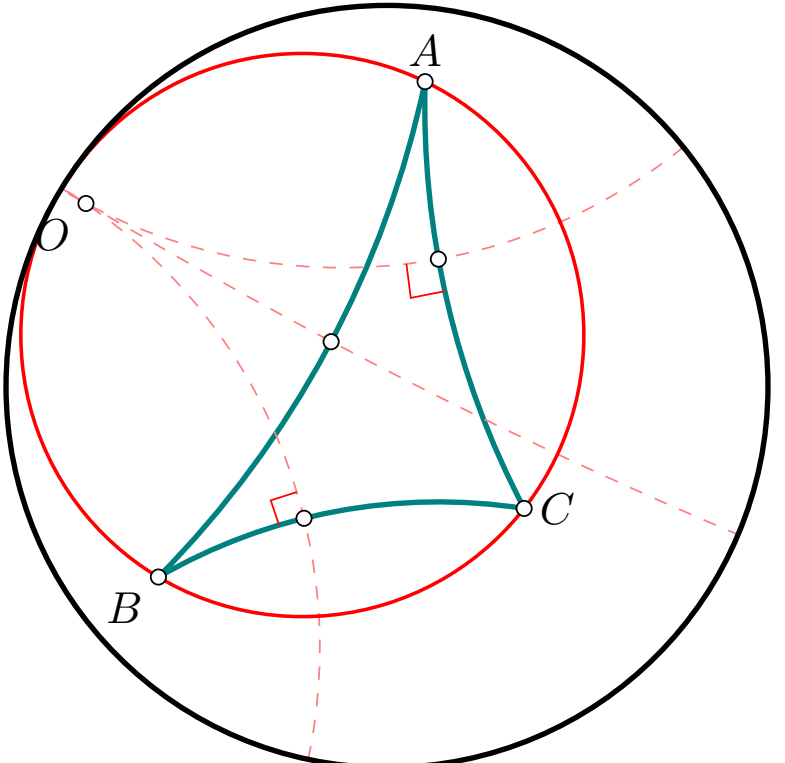
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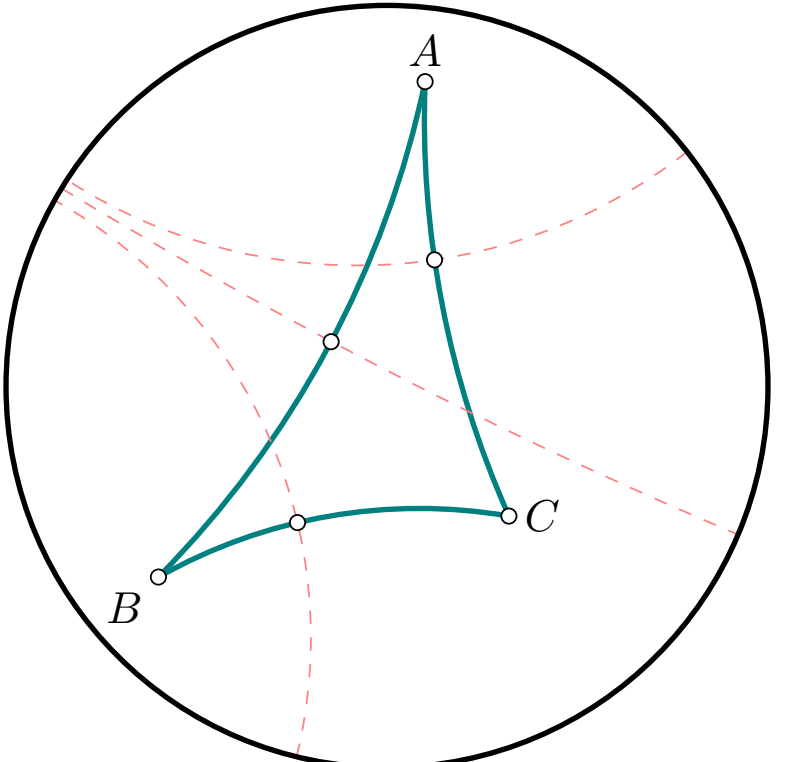
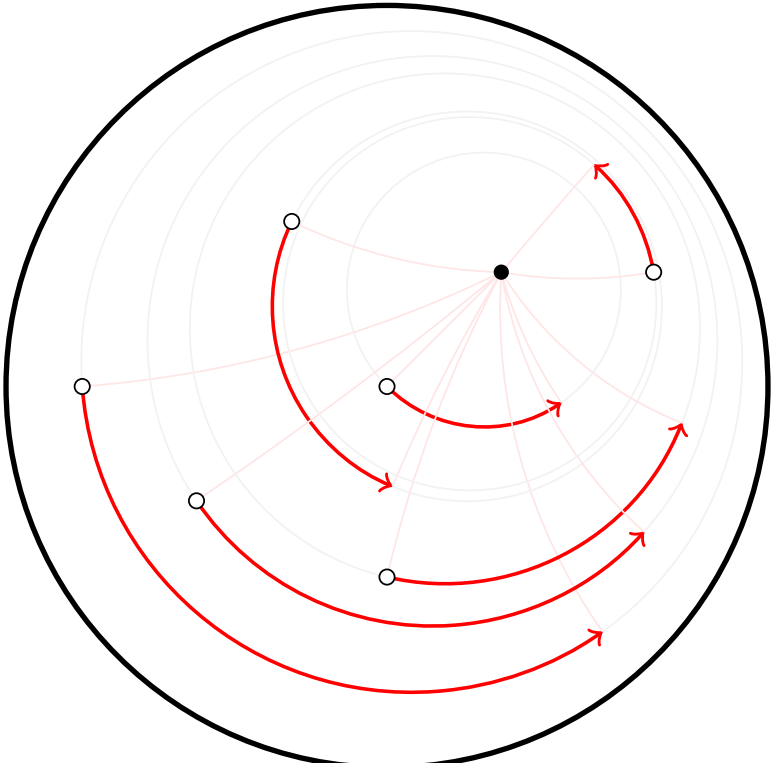
B
C

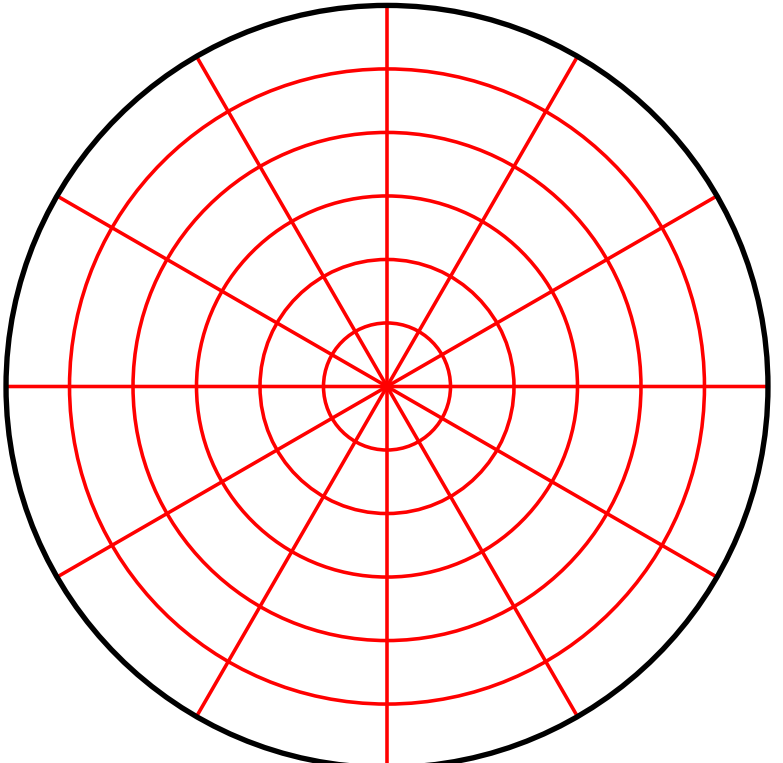

A
B
C

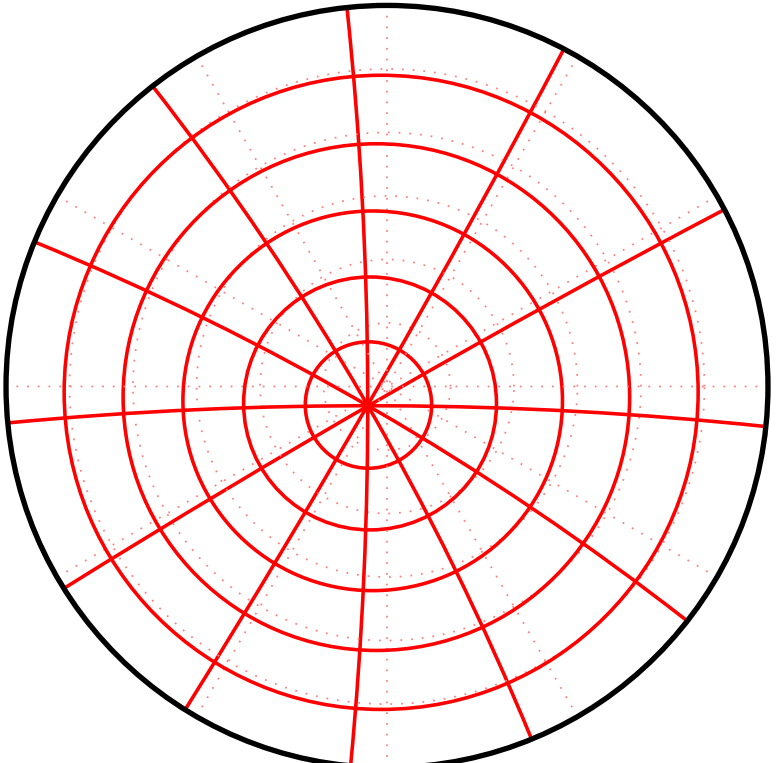
Transformations (isométries) du plan hyperbolique

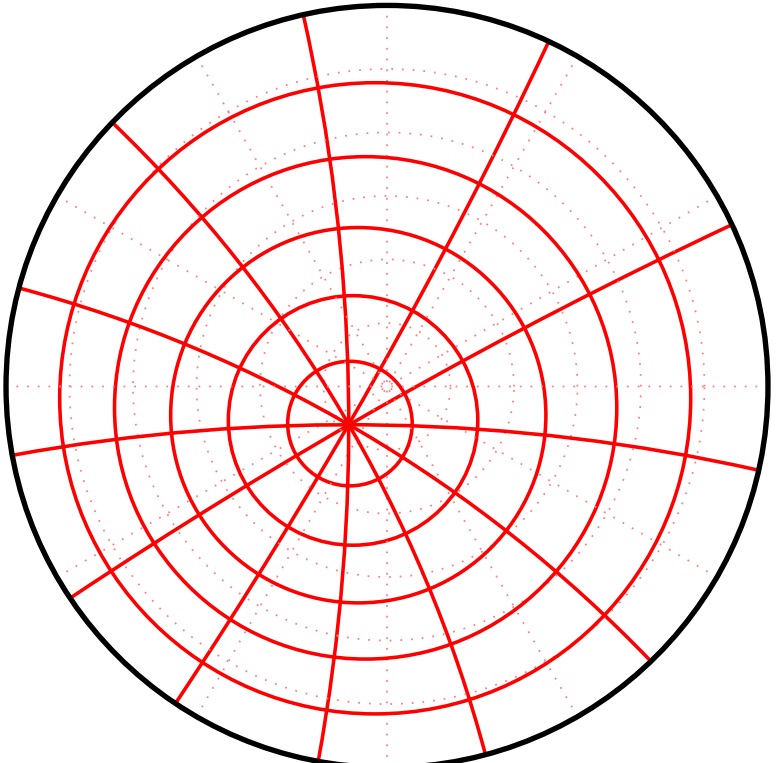
Automorphismes elliptiques (Analogue des rotations)

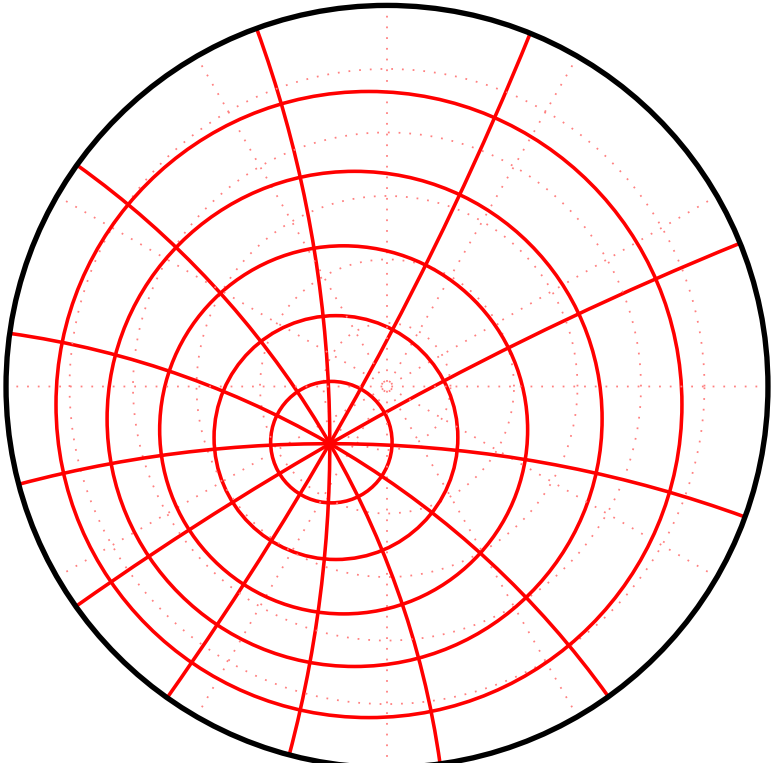


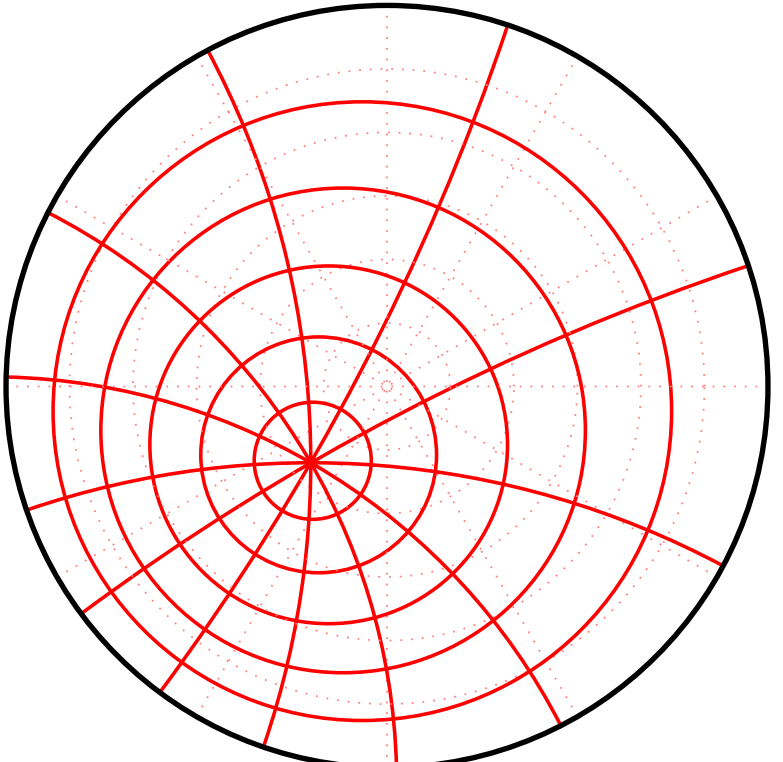
Automorphismes hyperboliques (Analogue des translations)

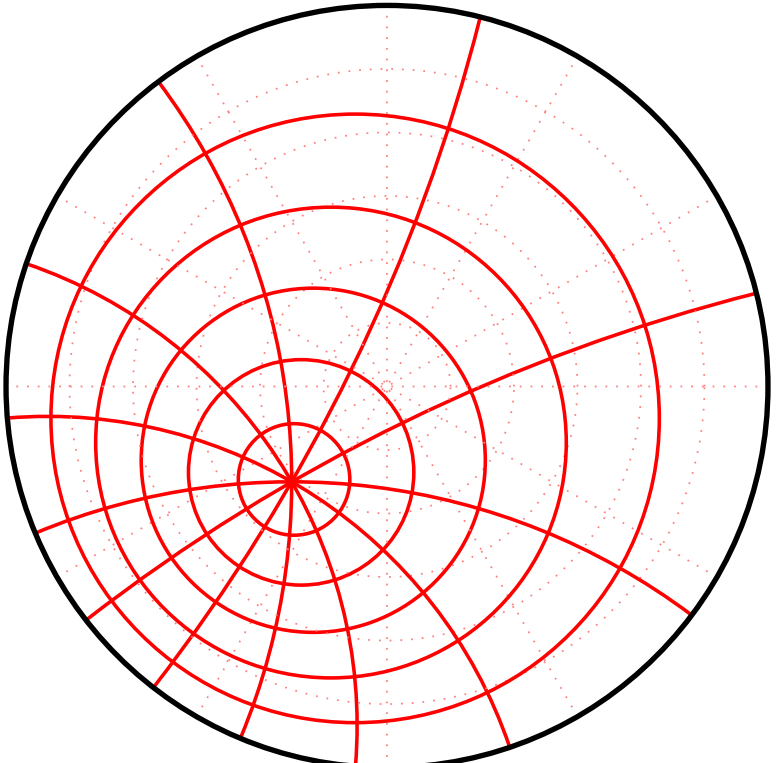


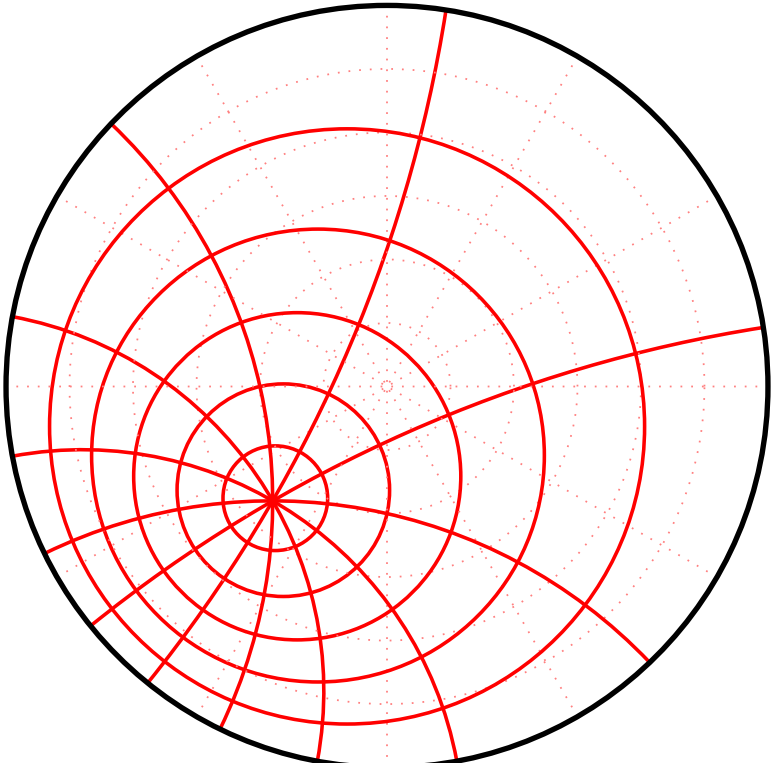


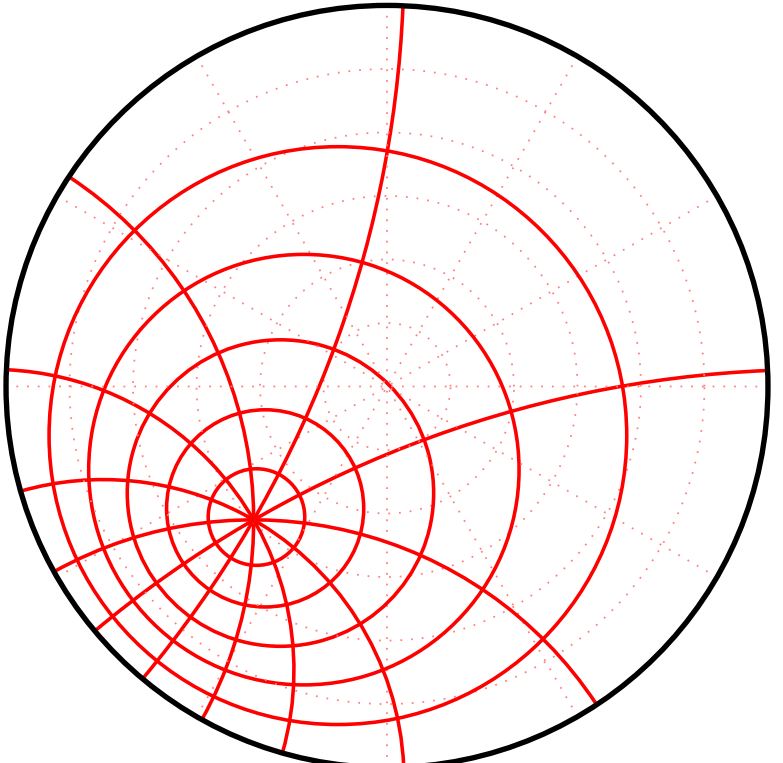


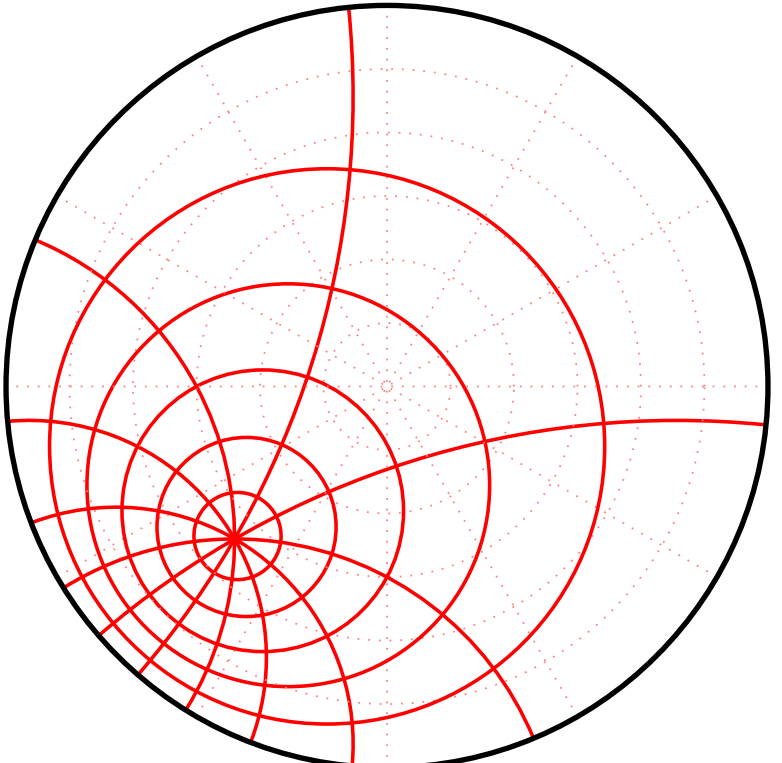


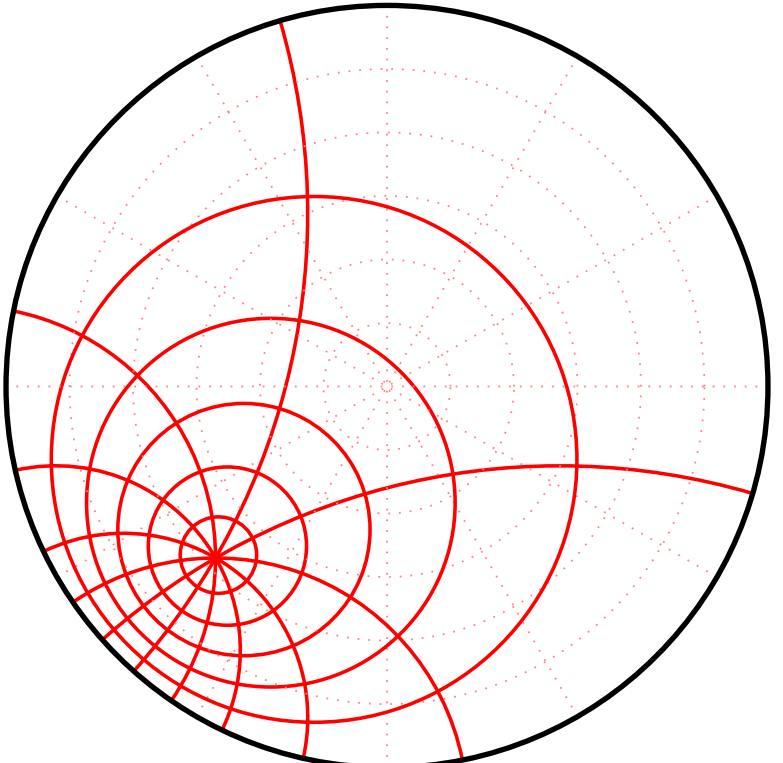


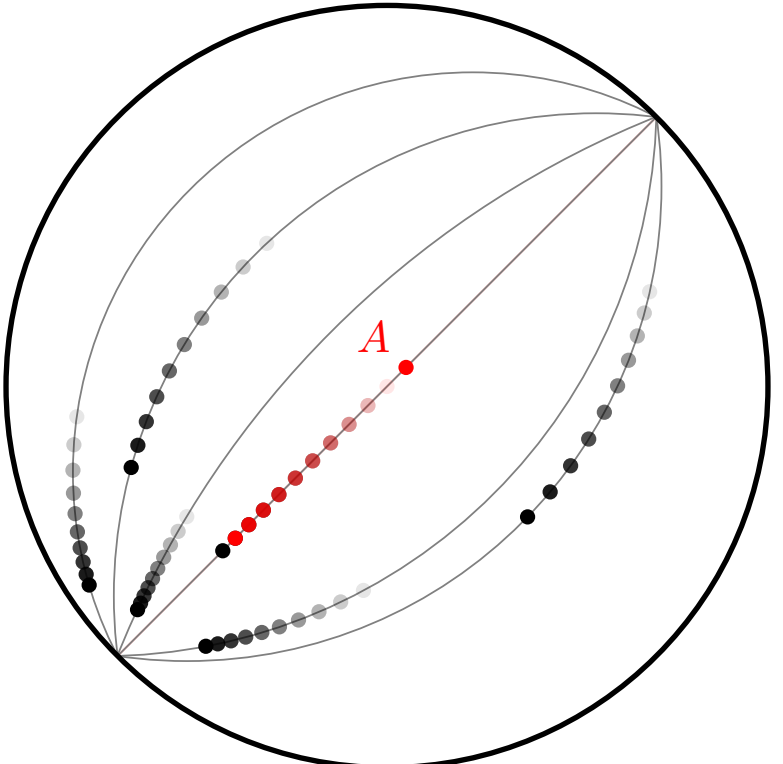






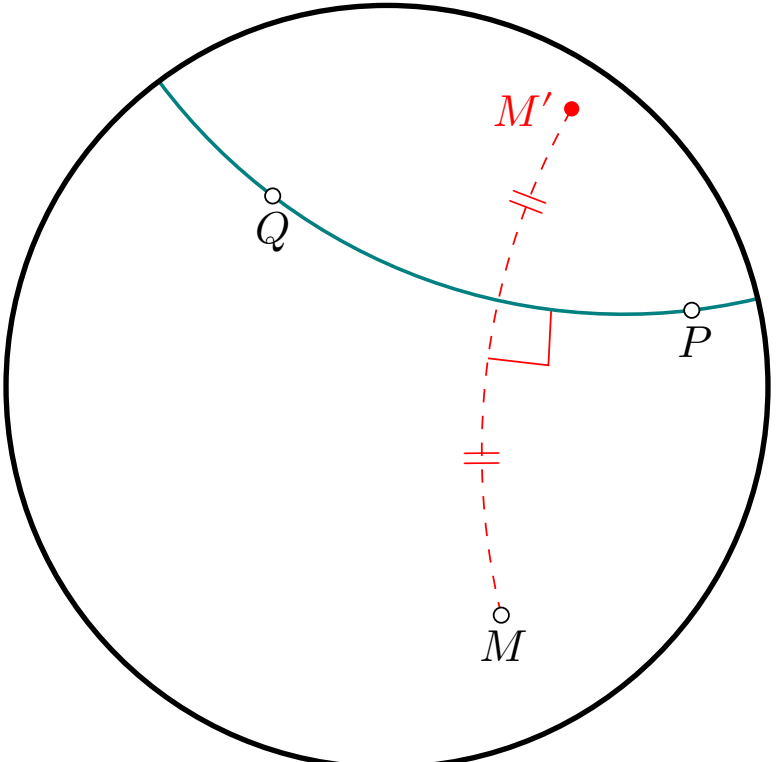


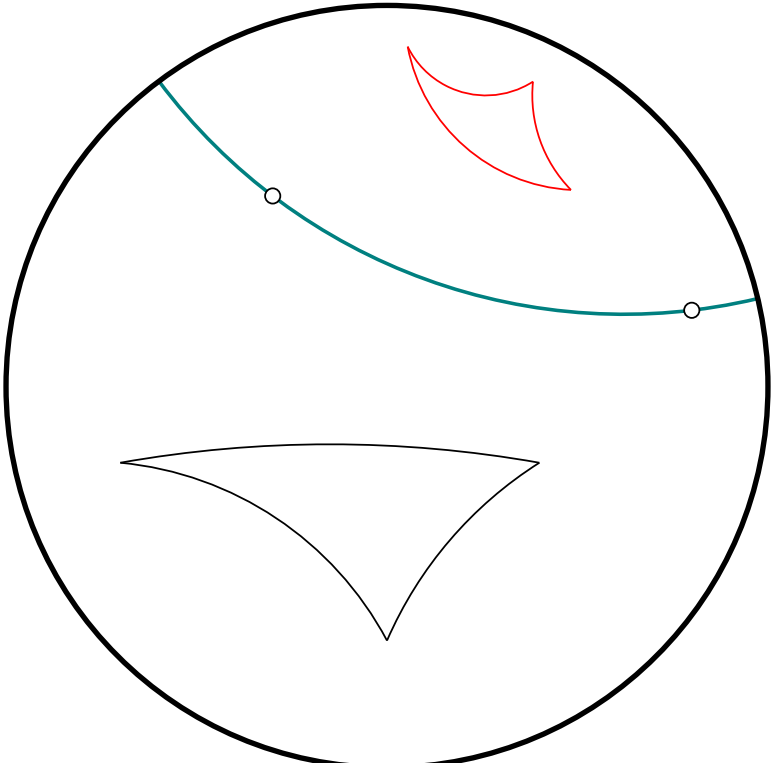



A

Automorphismes paraboliques (NON)

Symétries axiales


M'
Q
P
M



Partie II
Programmer en Lua dans un document (Lua)LaTeX

Compilation avec lualatex (plutôt que pdflatex)

Très simple : `lualatex myfile.tex`, ou `latexmk --lualatex myfile.tex`.
Preambule minimal :

```
\documentclass{article}
\usepackage[french]{babel}
\usepackage{mathtools} % à charger avant unicode-math
\usepackage{unicode-math}  % charge fontspec
```

Avantage immédiat (?) de la compilation avec LuaLaTeX choix de n'importe quelle police :

```
\setmainfont{Comic Sans MS}
Quelle indignité...
\setmainfont{Papyrus}
Je vous demande de vous arrêter !
```

Pour plus d'informations sur les polices disponibles, voir les documents récents de Daniel Flipo :
— `https://danielflipo.fr/doc/luatex/pdf2lua.pdf`
— `https://danielflipo.fr/doc/luatex/fonts.pdf`

Lua, un langage minimaliste (Mais efficace !)

— Très simple à apprendre
— Syntaxe très proche de Python (avec des `end` pour fermer les blocs)
— site web : `lua.org`
— Manuel : `https://www.lua.org/manual/`
— Livre : *Programming in Lua* (5ème édition, 1ère édition gratuite sur `https://www.lua.org/pil/contents.html`
— Notes de cours : `https://ic.ufrj.br/~fabiom/lua/`
— Utilisé dans un nombre croissant de packages latex, par exemple l'excellent `tkz-euclide`, pour augmenter la rapidité et la précision des calculs.

Utilisation de Lua dans un document LaTeX

```
La factorielle de $5$ vaut
\directlua{
  local result = 1
  for k=1,5 do result = result * k end
  tex.sprint(result)}.\par
La double factorielle de $5$,
notée $5!!$, est égale à
\directlua{local n, result = 5, 1
  while n>0 do
    result = result * n
    n = n-2
  end
  tex.sprint(result)}.\par
La puissance de $2$ immédiatement
(strict°) supérieure à $100$ est
\directlua{local n=1
  while n <= 100 do n = n * 2 end
  tex.sprint(n)}.
```

La factorielle de 5 vaut 120.
La double factorielle de 5, notée 5!!, est égale à 15.
La puissance de 2 immédiatement (strict°) supérieure à 100 est 128.

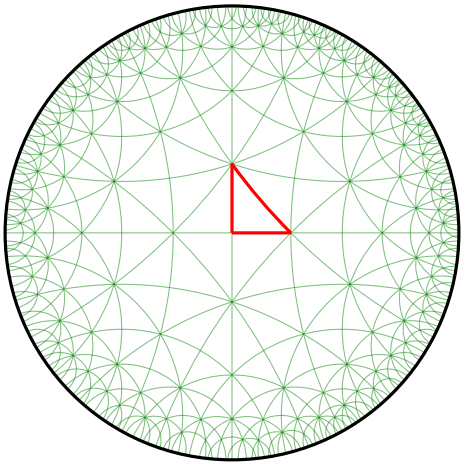
Partie III
Quelques fonctionnalités du package ‘luahyperbolic’

Informations pratiques et liens

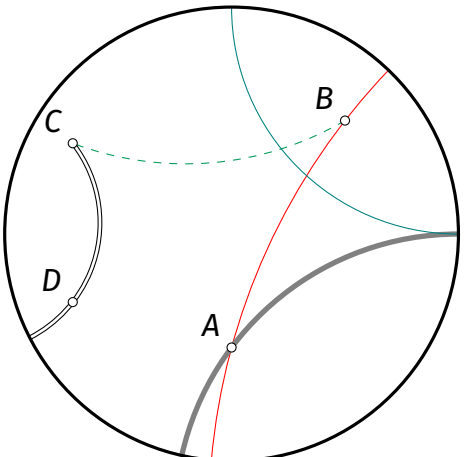
— Page CTAN : `https://ctan.org/pkg/luahyperbolic`

— Repo github : `https://github.com/dmegy/luahyperbolic`

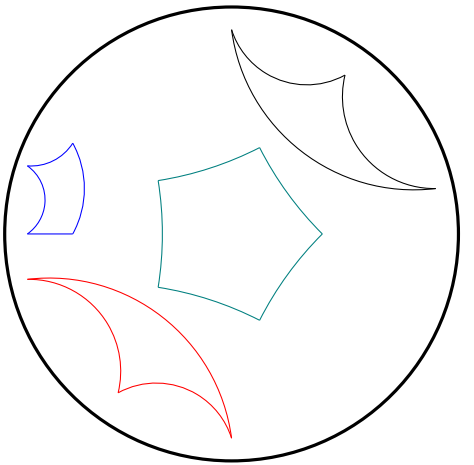
Attention, la version sur github est la plus à jour si l'on désire tester de nouvelles fonctionnalités.



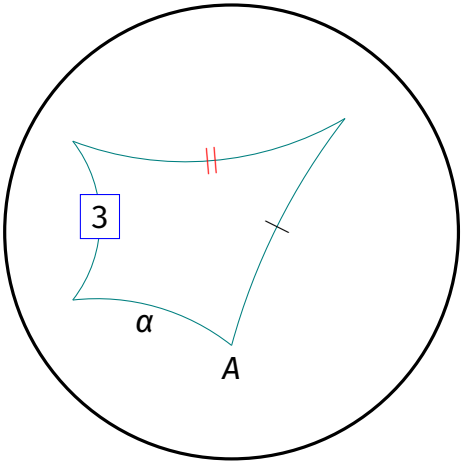
```
\begin{luacode*}
local A = -complex.I/2
local B = (1+complex.I)/2
local C = complex(-0.7,0.4)
local D = complex(-0.7,-0.3)
hyper.tikzBegin()
hyper.drawLine(A, B, "red")
hyper.drawSegment(B,C,"dashed, ForestGreen"
    ↪ )
hyper.drawRay(C,D,"double,double distance=1
    ↪ pt")
hyper.drawLine(A,1,"gray,line width=2pt")
hyper.drawLine(1, complex.I, "teal")
hyper.drawPoints(A, B, C, D)
hyper.labelPoints(A, "$A$", B, "$B$", C, "
    ↪ $C$", D, "$D$")
hyper.tikzEnd()
\end{luacode*}
```



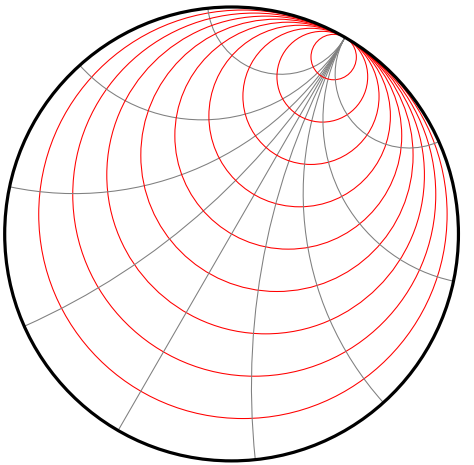
```
\begin{luacode*}
local A = complex(.9,.2)
local B = complex(.5,.7)
local C = complex(0,.9)
hyper.tikzBegin()
hyper.drawTriangle(A, B, C)
hyper.drawTriangle(-A, -B, -C, "red")
hyper.drawPolygon(-.9, -.7, complex(-.7,.4)
   ↪ , complex(-.9,.3), "blue")
local points={}
for k=1,5 do
  table.insert(points, complex.polar(.4,k
     ↪ *2*math.pi/5))
end
hyper.drawPolygonFromTable(points,"teal")
hyper.tikzEnd()
\end{luacode*}
```



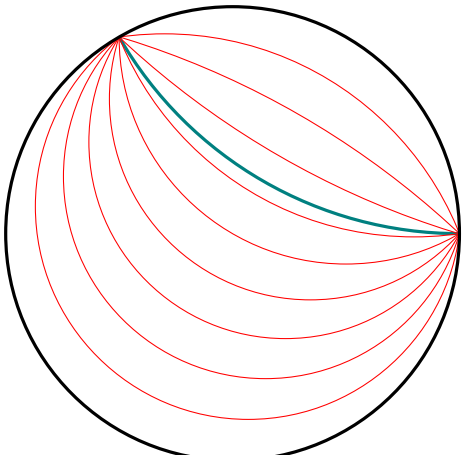
```
\begin{luacode*}
local A = -complex.I/2
local B = (1+complex.I)/2
local C = complex(-0.7,0.4)
local D = complex(-0.7,-0.3)
hyper.tikzBegin()
hyper.drawPolygon(A, B, C, D, "teal")
hyper.markSegment(A,B,"|")
local marking = "\\textcolor{red}{||}"
hyper.markSegment(B,C,marking)
hyper.labelSegment(C,D,"$3$", "fill=white,
   ↪ draw=blue")
hyper.labelSegment(D, A,"$\\alpha$", "below
   ↪ ")
hyper.labelPoint(A, "$A$", "below")
hyper.tikzEnd()
\end{luacode*}
```



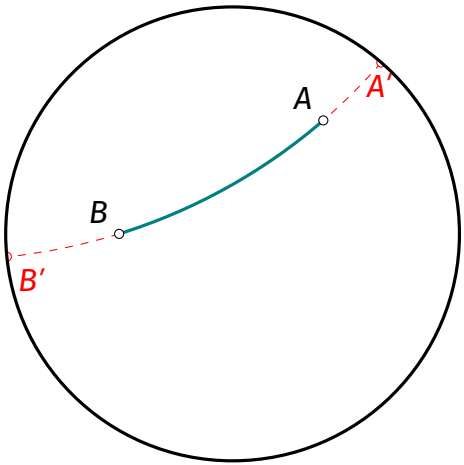
```
\begin{luacode*}
local theta=math.pi/3
local idealPoint = complex.exp_i(theta)
local N = 10
hyper.tikzBegin()
for k=1,N-1 do
  local point = idealPoint * (2*k/N-1)
  hyper.drawHorocycle(idealPoint,point,"red
    ↪ ")
  -- and an orthogonal geodesic :
  local endPoint = complex.exp_i(theta+2*k*
    ↪ math.pi/N)
  hyper.drawLine(idealPoint,endPoint,"gray"
    ↪ )
end
hyper.tikzEnd()
\end{luacode*}
```



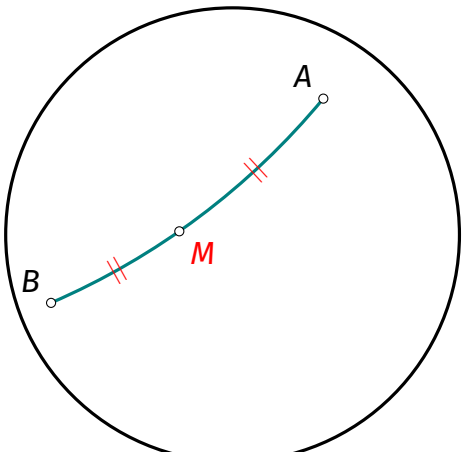
```
\begin{luacode*}
hyper.tikzBegin()
local A = 1
local B = complex.J
hyper.drawLine(A,B,"very thick, teal")
local N = 10
for k=1,N-1 do
  local P = (2*k/N-1)*(-B^2)
  hyper.drawHypercycleThrough(P, A, B, "red
    ↪ ")
end
hyper.tikzEnd()
\end{luacode*}
```



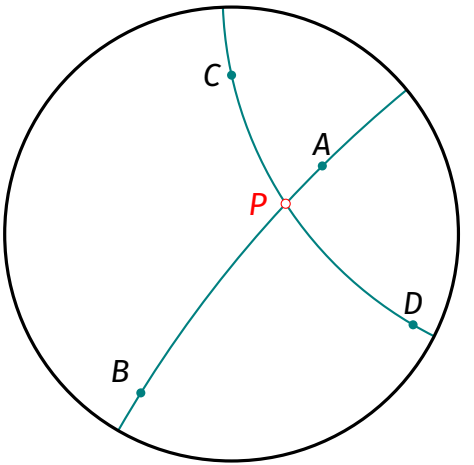
```
\begin{luacode*}
local A = complex(.4,.5)
local B = complex(-.5,0)
local AA, BB = hyper.endpoints(A,B)
hyper.tikzBegin()
hyper.drawSegment(A,B, "very thick, teal")
hyper.drawSegments(AA,A, B, BB, "dashed,
   ↪ red")
hyper.labelPoints(A, "$A$", B, "$B$")
hyper.labelPoint(AA, "$A'$", "red, below")
hyper.labelPoint(BB, "$B'$", "red, below
   ↪ right")
hyper.drawPoints(A, B)
hyper.drawPoints(AA,BB,"white, draw=red")
hyper.tikzEnd()
\end{luacode*}
```



```
\begin{luacode*}
local A = complex(0.4,0.6)
local B = complex(-0.8,-0.3)
local M = hyper.midpoint(A,B)
hyper.tikzBegin()
hyper.drawSegment(A,B, "very thick, teal")
hyper.drawPoints(A, B, M)
hyper.labelPoints(A, "$A$", B, "$B$")
hyper.labelPoint(M, "$M$", "below right,
   ↪ red")
local marking = "\\textcolor{red}{||}"
hyper.markSegment(A,M,marking)
hyper.markSegment(M,B,marking)
hyper.tikzEnd()
\end{luacode*}
```

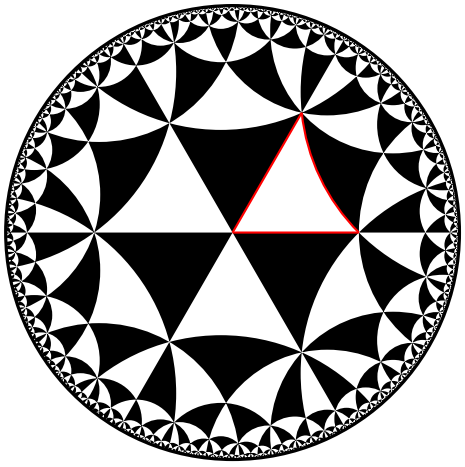


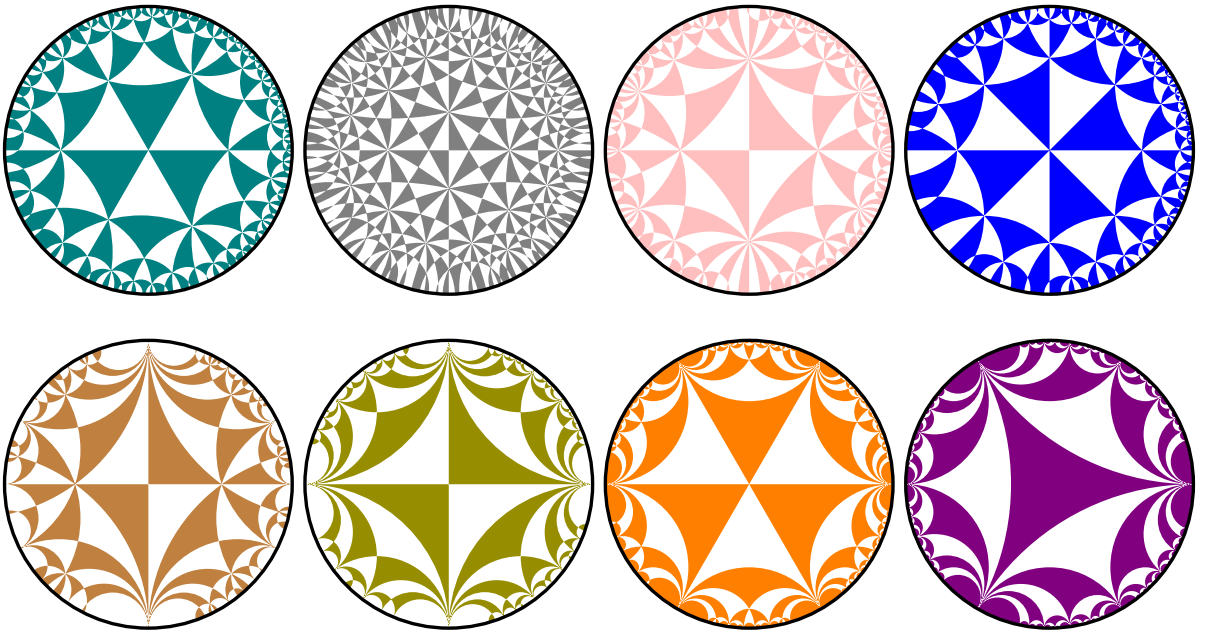
```
\begin{luacode*}
local A = complex(0.4,0.3)
local B = complex(-0.4,-0.7)
local C = complex(0,0.7)
local D = complex(0.8,-0.4)
local P = hyper.interLL(A, B, C, D)
hyper.tikzBegin()
hyper.drawLines(A, B, C, D, "thick, teal")
hyper.drawPoints(A, B, C, D,"thick, teal")
hyper.drawPoint(P,"white, draw=red")
hyper.labelPoint(A, "$A$", "above")
hyper.labelPoint(B, "$B$")
hyper.labelPoint(C, "$C$", "left")
hyper.labelPoint(D, "$D$", "above")
hyper.labelPoint(P, "$P$", "left=.1cm, red"
   ↪ )
hyper.tikzEnd()
\end{luacode*}
```



Pavages hyperboliques

```
\begin{luacode*}
local PI = math.pi
local DEPTH = 10
local A, B, C = hyper.triangleWithAngles(PI
    ↪ /3, PI/4, PI/5)
hyper.tikzBegin()
hyper.fillTilingFromTriangle(A, B, C, DEPTH
    ↪ )
-- default color is black
hyper.drawPolygon(A, B, C, "thick,red")
hyper.tikzEnd()
\end{luacode*}
```





Aperçu des fonctions disponibles

Module `complex' (petite lib de nombres complexes)

__add	abs	dot	isInteger	nonzero
__div	abs2	exp	isNear	polar
__eq	arg	exp_i	isNearInteger	rotate
__mul	clone	invert	isNot	rotate90
__pow	coerce	isClose	isReal	scal
__sub	conj	isColinear	isUnit	toPolar
__tostring	det	isComplex	log	unit
__unm	distinct	isImag	new	

Module `hyper' (lib principale)

Attention, les fonctions préfixées par « _ » vont peut-être redevenir privées donc non disponibles pour l'utilisateur. Il est demandé de ne pas les utiliser.

_assert
_assert_in_closed_disk
_assert_in_disk
_circle_to_euclidean
_coerce_assert_in_closed_disk
_coerce_assert_in_disk
_ensure_order
_error
_geodesic_data
_in_closed_disk
_in_disk
_in_half_plane
_metric_factor
_midpoint_to_origin
_on_circle
_on_line
_on_segment
_radial_half
_radial_scale
_warning
angle
automorphism
automorphismFromPairs
automorphismSending
barycenter2
distance

distanceToLine
drawAngle
drawAngleBisector
drawArc
drawCircle
drawCircleThrough
drawCircumcircle
drawHorocycle
drawHypercycleThrough
drawIncircle
drawLine
drawLineFromVector
drawLines
drawLinesFromTable
drawPerpendicularBisector
drawPerpendicularThrough
drawPoint
drawPointOrbit
drawPoints
drawPolygon
drawPolygonFromTable
drawPolyline
drawPolylineFromTable
drawRay
drawRayFromPoints
drawRayFromVector
drawRegularPolygon
drawRightAngle
drawSegment
drawSegments
drawSemicircle
drawTangentAt
drawTilingFromAngles
drawTilingFromTriangle

drawTriangle
drawVector
endpoints
endpointsAngleBisector
endpointsPerpendicularBisector
expMap
fillCircle
fillTilingFromAngles
fillTilingFromTriangle
interCC
interLC
interLL
interpolate
labelPoint
labelPoints
labelSegment
markSegment
markSegments
midpoint
mobiusTransformation
parabolic
pointOrbit
projection
propagateGeodesics
randomPoint
reflection
rotation
rotationFromPair
symmetry
symmetryAroundMidpoint
tangentVector
tikzBegin
tikzClearBuffer
tikzDefineNodes

tikzEnd
tikzExport
tikzGetFirstLines
tikzPrintf
tikz_shape_closed_line
tikz_shape_euclidean_segment
tikz_shape_segment
triangleCentroid
triangleCircumcenter
triangleIncenter
triangleOrthocenter
triangleWithAngles

Merci !

Dernière version et code source des slides :

`https://github.com/dmegy/expose-luahyperbolic-gutenberg-2026`